

Market Counterfactuals with Nonparametric Supply: An ML/AI Approach^{*}

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Abstract

We develop a new approach to market counterfactuals (e.g., merger simulation, tax policy, and product regulation) using machine learning and nonparametric structure from economics. Building on [Berry and Haile \(2014\)](#), we propose a flexible supply specification that relaxes restrictive assumptions about firm conduct and costs. We adapt the Variational Method of Moments ([Bennett and Kallus, 2023](#)) with deep neural networks to estimate the model, addressing endogeneity through instrumental variables. Monte Carlo evidence demonstrates good performance even in high-dimensional environments and with moderate sample sizes. Applied to the American Airlines-US Airways merger, our method achieves a fivefold reduction in price prediction error versus standard Bertrand-Nash models.

KEYWORDS: counterfactual analysis, variational method of moments (VMM), neural networks, merger simulation, airline markets

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1 Introduction

The responses of consumers and firms mediate the effects of policy or market design interventions. Designing good policy, therefore, requires an assessment of what would happen under different *counterfactual* policy scenarios, possibly never observed before. Economists routinely assess market counterfactuals to evaluate policy (prospectively or retrospectively) in a variety of domains. Examples include the evaluation of mergers (e.g., [Miller and Weinberg, 2017](#)), trade policy (e.g., [Berry, Levinsohn, and Pakes, 1999](#)), environmental policy (e.g., [Barwick, Kwon, and Li, 2024](#)), tax (e.g., [Miravete, Seim, and Thurk, 2018](#)) and non-tax (e.g., [Barahona, Otero, and Otero, 2023](#)) interventions in markets with externalities, the design of healthcare markets (e.g., [Tebaldi, 2025](#)), vouchers and other interventions in education markets (e.g., [Neilson, 2025](#)), and regulation of financial markets (e.g., [Bhattacharya and Illanes, 2025](#)).

The standard approach is to perform these counterfactuals using parametric demand and supply models to capture consumers’ and firms’ responses to policy changes. While recent advances have introduced more flexibility on the demand side (e.g., [Compiani, 2022](#)), supply-side modeling remains highly constrained by parametric assumptions about firms’ cost and the nature of competition, often defaulting to models of Bertrand pricing and constant marginal cost. These restrictive assumptions may lead to misleading predictions when the true model of firm conduct and cost differs from the imposed structure. This is not only a theoretical problem: for instance, evidence from merger retrospectives (e.g., [Peters, 2006](#); [Björnerstedt and Verboven, 2016](#); [Bhattacharya, Kreps, Illanes, Salas, and Stillerman, 2025](#)) highlights the importance of supply-side assumptions and the potential shortcomings of standard approaches.

This paper proposes a new approach that maintains the basic economic insight that counterfactual outcomes arise in equilibrium, while relaxing parametric assumptions on the supply side. Building on [Berry and Haile \(2014\)](#), we introduce a *flexible supply function* that combines markups and marginal costs into a single object estimated without specifying the model of competition. Equilibrium prices depend nonparametrically on market shares, demand derivatives, cost shifters, and ownership structure, nesting standard oligopoly models while letting the data reveal how firms set prices. Demand is assumed to be known or estimated in a first step. Although the resulting model is high-dimensional (with J products, supply depends on $J + J^2$ endogenous arguments), identification is feasible with standard IV variation avail-

able in typical differentiated-product settings: demand shifters and rival cost shifters provide the needed exogenous variation, because knowledge of the demand structure links shares and demand derivatives, reducing the effective dimensionality of the identification problem. Symmetry restrictions can further assist with identification, given the variation within markets.

We estimate the high-dimensional supply function using the neural Variational Method of Moments (VMM) of [Bennett and Kallus \(2023\)](#), which reformulates the IV problem as a minimax optimization with neural networks. VMM coincides with optimally weighted Generalized Method of Moments (GMM) in parametric settings, making it a natural generalization of standard structural methods. In our nonparametric setting, deep neural networks can exploit the compositional structure inherent in oligopoly pricing (e.g., [Bauer and Kohler, 2019](#); [Schmidt-Hieber, 2020](#)), achieving faster convergence rates than traditional nonparametric IV methods. We also develop inference procedures for counterfactual predictions, extending [Bennett and Kallus \(2023\)](#) to construct simultaneous confidence intervals via the numerical delta method and Holm’s step-down procedure. The resulting toolkit completes estimation within hours on standard hardware for typical IO datasets.

In essence, our method combines domain knowledge from economics (which guides the formulation of the nonparametric model and the choice of instruments, encoded in the moment condition) with a data-driven ML/AI procedure (implemented with neural VMM). A central contribution of this paper is to demonstrate that this toolkit works in practice. We validate this claim along two dimensions: Monte Carlo simulations that stress-test the method across a range of environments, and an empirical application to airline mergers that demonstrates the method’s value with real data.

Our Monte Carlo simulations showcase the method’s performance across four dimensions. First, we demonstrate predictive accuracy: in hold-out samples, the flexible model achieves mean squared errors (MSE) close to those of correctly specified models with just 100 training markets, while misspecified parametric models generate errors 3–6 times larger. Second, we establish scalability to high-dimensional environments with 30 products similar to [Miller and Weinberg \(2017\)](#), where the inclusion of demand derivatives and larger network architectures proves crucial. Third, we verify economic interpretability by showing that the flexible model precisely recovers underlying pass-through matrices, including features like negative cross-price effects under partial internalization that misspecified models would miss, demonstrating that

our approach captures the true economic structure without imposing it. Fourth, we validate counterfactual prediction in policy-relevant scenarios: regulations pushing product characteristics outside the training support, Laffer curves extrapolating to tax levels triple those in the training data, and merger simulations (where we obtain consumer surplus prediction errors less than half those of misspecified models). Finally, our inference procedure delivers reliable confidence intervals with 94–98% coverage rates in samples of 1,000 markets.

Besides standard parametric approaches, a natural alternative to our method is to bypass the structural supply function entirely and train a flexible machine-learning model to predict prices directly from exogenous variables. This reduced-form approach is appealing because it sidesteps the need for instruments and equilibrium conditioning, but [Berry and Benkard \(2006\)](#) show that the reduced-form price mapping is not nonparametrically identified in differentiated-product markets. Our simulations confirm this empirically: a neural-network predictor trained only on exogenous variables, rather than on the equilibrium objects entering the structural supply function, performs poorly in counterfactuals.

We also apply our method to the 2013 American Airlines-US Airways merger. Focusing on markets that transitioned from three to two airlines, the flexible model achieves a 44% improvement in pre-merger fit over standard Bertrand-Nash and a fivefold reduction in passenger-weighted mean squared error for post-merger price predictions. While the Bertrand-Nash model systematically overpredicts price increases, echoing results in [Bhattacharya et al. \(2025\)](#), our flexible approach centers its predictions around observed outcomes with a fivefold reduction in passenger-weighted mean squared error. The improved accuracy is a result of the model’s ability to learn actual competitive conduct from the data rather than imposing it, highlighting the practical importance of relaxing standard assumptions in merger evaluation.

Although the approach we propose offers significant advantages in flexibility, it involves trade-offs that should guide its application. Two dimensions are particularly relevant. The first is the exogenous variation available in the data: as a nonparametric method, our approach requires richer identifying variation than conduct testing ([Backus, Conlon, and Sinkinson, 2021](#); [Duarte, Magnolfi, Sølvesten, and Sullivan, 2024](#)) or more parametric supply approaches ([Magnolfi and Sullivan, 2022](#)). The second is extrapolation: parametric models extrapolate freely because functional form assumptions pin down behavior everywhere; our approach is in principle less reliable

far from the training support, though our simulations demonstrate robustness well beyond it. On the other hand, our method imposes no functional form on the supply side, carrying the lowest misspecification risk. How this mix of advantages and limitations plays out depends on the application, which is why we view our method as complementary to existing parametric and testing approaches.

This paper contributes to the IO literature that proposes nonparametric models of market equilibrium. We build our flexible model of supply on the identification results of [Berry and Haile \(2014\)](#), obtaining new results on the nonparametric identification of supply. Similarly to [Compiani \(2022\)](#), who develops a method to estimate demand nonparametrically, our paper proposes a method for nonparametric estimation of the supply side. With similar motivation, [Gandhi and Houde \(2020a\)](#) (proposing a linear approximation of the markup function) and [Otsu and Pesendorfer \(2024\)](#) (using the revelation principle) also develop methods to bring more flexible supply models to data.¹ We complement these approaches by proposing a method that leverages advances in ML/AI coupled with a nonparametric structure.

A growing literature applies ML/AI methods in economics beyond pure prediction, including double/debiased machine learning (e.g., [Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins, 2018](#)) and causal forests (e.g., [Athey and Wager, 2018](#)). In structural estimation, [Kaji, Manresa, and Pouliot \(2023\)](#) introduce adversarial estimation using neural network discriminators, and [Wei and Jiang \(2025\)](#) train neural networks to map data moments to structural parameters. Our estimation problem is a conditional moment restriction, placing it within the strand of this literature that applies neural networks to nonparametric IV.

The use of neural networks as nonparametric sieve estimators has a long history in econometrics ([Chen and White, 1999](#); [Ai and Chen, 2007](#)). More recently, [Hartford, Lewis, Leyton-Brown, and Taddy \(2017\)](#) pioneer a two-stage deep learning approach to NPIV, while [Dikkala, Lewis, Mackey, and Syrgkanis \(2020\)](#) reformulate IV estimation as adversarial optimization over moment conditions, and [Chen, Chen, and Tamer \(2023\)](#) and [Chen, Liao, and Wang \(2025a\)](#) use deep neural networks as non-linear sieves, obtaining semiparametric efficiency for average derivatives of NPIV and NPQIV models.² We adopt the neural VMM of [Bennett and Kallus \(2023\)](#) because it

¹Nonparametric structural methods have also been fruitfully applied to production function estimation; see, e.g., [Akerberg, Chen, Hahn, and Liao \(2014\)](#).

²See [Wu, Kuang, Xiong, and Wu \(2025\)](#) for a survey of ML/AI approaches to IV estimation.

is based on conditional moment conditions, coincides with optimally weighted GMM in parametric settings, and attains semiparametric efficiency.

The remainder of the paper is organized as follows. Section 2 presents the standard framework that serves as our benchmark. Section 3 develops our flexible approach and discusses identification. Section 4 describes the VMM estimation procedure and its properties. Section 5 presents Monte Carlo evidence, and Section 6 applies our methodology to airline mergers. Section 7 concludes.

2 Market Equilibria and Counterfactuals

This section presents a general equilibrium model for differentiated product markets, following Berry and Haile (2014). We describe the data-generating process, define market counterfactuals, and illustrate with merger simulation as a concrete example.

Data-generating Process: Consumers choose products j from a set $\mathcal{J} = \{1, \dots, J\}$ offered by firms across markets $t \in \mathcal{T} = \{1, \dots, T\}$.³ Each market has a measure M_t of consumers, which we call market size, and a $J \times J$ ownership matrix defined as $\mathcal{H}_t = [h_{jkt}]$ with $h_{jkt} = 1$ if the same firm owns products j and k . Each product-market pair (j, t) has a price $p_{jt} \in \mathbb{R}$, a market share $s_{jt} \in (0, 1)$, a vector of product characteristics $x_{jt} \in \mathbb{R}^K$ that enter consumers' demand, and a vector of cost shifters $w_{jt} \in \mathbb{R}^L$. To streamline notation, for any variable y_{jt} , we denote y_t as the vector of values in market t . Endogenous prices and quantities are generated by the equilibrium of demand and supply in market t ;⁴ we describe these in turn.

Across all markets, the demand system $\mathcal{s}(\cdot) = (\mathcal{s}_1(\cdot), \dots, \mathcal{s}_J(\cdot))$ is given by:

$$s_{jt} = \mathcal{s}_j(p_t, x_t, \xi_t), \quad j = 1, \dots, J$$

where $\xi_t = (\xi_{1t}, \dots, \xi_{Jt})'$ is a vector of unobservable product characteristics. Define the corresponding matrix of demand derivatives as:

$$D_t \equiv \frac{\partial \mathcal{s}(p_t, x_t, \xi_t)}{\partial p'_t} = \left[\frac{\partial s_{kt}}{\partial p_{jt}} \right]_{j,k=1}^J.$$

³For expositional simplicity, we keep the set of products offered constant across markets, but none of our results will depend on this assumption.

⁴The environment can be extended to include other endogenous variables, e.g., product quality.

In general, these derivatives are a function of equilibrium outcomes and exogenous product characteristics, or $D_t = D(p_t, x_t, \xi_t)$.

On the supply side, firm behavior is characterized by a system of first-order conditions for the firms' profit maximization problems:

$$p_{jt} = \Delta_{jt} + c_{jt}, \quad j = 1, \dots, J,$$

where, for each product j , Δ_{jt} is the markup, and c_{jt} is the marginal cost. Markups, which can be expressed as functions $\Delta_{jt} = \Delta_j(s_t, p_t, D(p_t, x_t, \xi_t), \mathcal{H}_t)$, arise endogenously from a model of firm conduct. Firm j 's costs are generated by some cost function c_j , or $c_{jt} = c_j(q_t, w_{jt}, \omega_{jt})$, where q_{jt} and ω_{jt} are, respectively, equilibrium quantities (obtained as the product of market size and market share, or $q_{jt} = M_t s_{jt}$) and unobserved cost shifter variables. The dependence of costs on q_t allows for the presence of economies of scale and scope. Therefore, a markup function and a cost function are the key ingredients of the supply-side model.

Example 1. *A canonical assumption in the literature (e.g., in [Berry, Levinsohn, and Pakes, 1995](#)) is that the model of conduct is Bertrand-Nash price competition (hereafter Bertrand), whereby firms play a simultaneous pricing game, with constant marginal cost that is a linear index of cost shifters. Under this assumption, market-level markups are $\Delta_t = (\mathcal{H}_t \odot D_t)^{-1} s_t$, where $\odot$ denotes the Hadamard product. Furthermore, the cost function is often specified as linear: $c_t = w_t' \gamma + \omega_t$.*

To summarize the environment, we have described three functions that pin down the structure of demand ($\mathcal{J}(\cdot)$) and supply ($\Delta(\cdot)$, $c(\cdot)$). The endogenous outcomes (p_t, s_t) are generated by these primitives as a function of exogenous variables $(x_t, w_t, \xi_t, \omega_t)$. This formulation nests the standard Bertrand model but allows for more general forms of *static* oligopoly interaction. However, it rules out important supply settings without a static first-order-condition characterization, such as the price leadership equilibrium of [Miller, Sheu, and Weinberg \(2021\)](#).

We now restrict the environment with some assumptions on the primitives. We start with an assumption on equilibrium selection:

Assumption 1. (*Equilibrium Selection*) There exists a unique equilibrium, or the equilibrium selection rule is such that the same p_t arises whenever the vector $(x_t, w_t, \xi_t, \omega_t)$ is the same.

This assumption, similar to Assumption 13 in [Berry and Haile \(2014\)](#), ensures that prices reflect stable equilibrium behavior. We then impose a mild assumption on cost:

Assumption 2. (*Separability of Cost*) For any product $j = 1, \dots, J$, the cost function is separable in unobservable shocks, or $c_j(q_t, w_{jt}, \omega_{jt}) = \bar{c}_j(q_t, w_{jt}) + \omega_{jt}$.

This separability assumption is essentially without loss of generality because the unobservable component ω_{jt} can be defined as the residual between total costs and the component explained by observables.

We also require the known demand system to satisfy an index restriction on how product characteristics enter demand, which mirrors similar restrictions in [Berry and Haile \(2014\)](#). We partition the product characteristics as $x_{jt} = (x_{jt}^{(1)}, x_{jt}^{(2)})$, where $x_{jt}^{(1)} \in \mathbb{R}$ enters demand only through a linear index with the demand unobservable, while $x_{jt}^{(2)} \in \mathbb{R}^{K-1}$ may enter demand more flexibly.

Assumption 3. (*Index Structure for Demand*) The demand unobservable ξ_{jt} and the characteristic $x_{jt}^{(1)}$ enter preferences only through their sum $\delta_{jt} = x_{jt}^{(1)} + \xi_{jt}$. Therefore, demand, inverse demand, and demand derivatives can be written equivalently as functions of (ξ_t, x_t) or as functions of $(\delta_t, x_t^{(2)})$.

As we show in Section 3.2, this index structure also facilitates identification of supply by creating a lower-dimensional manifold on which the supply function must be identified. Finally, we restrict the class of supply models we consider, so that prices and product characteristics enter markups only through demand derivatives:

Assumption 4. (*Markup Dependence*) For any product $j = 1, \dots, J$, the markup function $\Delta_j(\cdot)$ depends on endogenous market shares s_t and on the matrix of demand derivatives D_t , but conditional on these variables, does not depend on prices p_t or product characteristics (x_t, ξ_t) .

Assumption 4 is satisfied by a broad range of conduct models beyond Bertrand, including Cournot competition, various forms of partial collusion, and models where firms maximize weighted combinations of profits and consumer surplus.⁵ Under this assumption we can write markup functions for any $j = 1, \dots, J$ as $\Delta_{jt} = \Delta_j(s_t, D_t, \mathcal{H}_t)$.

⁵For a thorough discussion of how the models above (and more) satisfy this assumption, see Appendix C in [Dearing, Magnolfi, Quint, Sullivan, and Waldfogel \(2024\)](#).

Observables for the Researcher: In line with standard market data environments, observable variables for the researcher include (s_t, p_t, x_t, w_t) , as well as M_t and $\mathcal{H}_t$. In addition, to identify our model of supply, we will assume that demand is identified, so that the researcher can identify ξ_t and D_t . This echoes standard “two-step procedures” where demand is first estimated or calibrated before estimation or testing of supply.

Assumption 5. (*Known Demand*) The matrix of demand derivatives is known, so that $D_t = D(p_t, x_t, \xi_t)$ is observed.

Assumption 5 will be maintained for our identification results in Section 3.2: the supply function h is identified given knowledge of D_t . In practice, demand is estimated in a first step (e.g., via BLP or maximum likelihood), and the researcher observes $\hat{D}_t$ rather than D_t . We discuss how to account for first-step estimation error in Section 4.

Market counterfactuals: Most policy changes of interest in applied research can be modeled as *market counterfactuals*. We will predict counterfactuals for a specific set of markets (periods) $t \in \tilde{\mathcal{T}}$ outside of the sample; here and in what follows, we use tildes to denote counterfactual objects.

A market counterfactual must involve a change in either demand or supply functions, or any of their exogenous arguments. For instance, exogenous product characteristics may now take values $\tilde{x}_{jt}$, or cost functions may be altered to $\tilde{c}_j(\cdot)$. Under the new primitives, consumers and firms react to exogenous changes and counterfactual market outcomes arise from a new market equilibrium $(\tilde{p}_t, \tilde{s}_t)$. These outcomes need to satisfy demand and supply equations, and under our assumptions, can be found as the fixed point of the system:

$$\tilde{p}_{jt} = \tilde{\Delta}_j(\tilde{p}_t, \tilde{D}(\tilde{p}_t, \tilde{x}_t, \tilde{\xi}_t), \tilde{\mathcal{H}}_t) + \tilde{c}_j(\tilde{q}_t, \tilde{w}_{jt}, \tilde{\omega}_{jt}), \quad j = 1, \dots, J.$$

For any market $t \in \tilde{\mathcal{T}}$, counterfactuals of interests can then be expressed as a map $F(\tilde{p}_t, \tilde{s}_t)$, where we suppress the dependence on (counterfactual) structural objects and exogenous variables. To illustrate the general notion of a market counterfactual, we consider the standard merger simulation problem.

Example 2 (Merger Simulation). *Suppose the researcher wants to predict the unilateral effect on prices of a horizontal merger. Further suppose that the true model of conduct is Bertrand; other assumptions are possible, including Cournot (e.g., [Peters](#),*

2006), Nash Bargaining (e.g., [Gowrisankaran, Nevo, and Town, 2015](#)), or profit-weight models (e.g., [Miller and Weinberg, 2017](#)). As a baseline, the merger results only in a deterministic change to the counterfactual ownership matrix $\tilde{\mathcal{H}}_t$ for market t . The map F could return predicted counterfactual $(\tilde{p}_t, \tilde{s}_t)$, or other objects of interest such as counterfactual consumer surplus and firm profit.

Estimating a Counterfactual: Evaluating the value of F for specific counterfactuals requires predicting endogenous counterfactual outcomes $(\tilde{p}_t, \tilde{s}_t)$, which in turn requires knowledge of the (counterfactual) primitives, exogenous observables, and unobservables. In practice, researchers use a combination of data and assumptions. Typically, the functions s and $\bar{c}$ are estimated with parametric models, and markup functions $\Delta_j(\cdot)$ follow from an assumption on conduct. This allows the researcher to obtain estimates of ξ_t and ω_t , which are used to specify counterfactual $\tilde{\xi}_t$ and $\tilde{\omega}_t$. Counterfactual $\tilde{j}(\cdot)$, $\tilde{c}(\cdot)$, and $\tilde{\Delta}(\cdot)$ are either kept fixed, or changed deterministically.

Example 2 (continued). *Standard merger simulation* (e.g., [Nevo, 2000](#)) typically proceeds in three steps. First, the researcher formulates a parametric (e.g., linear, logit, or mixed logit) demand system $s(\cdot; \theta^D)$ and estimates demand primitives $\hat{\theta}^D$, which imply estimates of $\hat{\xi}_t$ and demand derivatives $\hat{D}_t = D(p_t, x_t, \hat{\xi}_t; \hat{\theta}^D)$ in each market t . Second, the researcher assumes a supply model, which encompasses models of cost and markups. In the standard toolkit, these are typically a constant marginal cost function, and Bertrand conduct. Marginal costs can thus be inverted as $\hat{c}_t = p_t - (\mathcal{H}_t \odot \hat{D}_t)^{-1} s_t$. Third, the researcher computes post-merger prices $\tilde{p}_t$ in each market t under the post-merger ownership structure $\tilde{\mathcal{H}}_t$ holding everything else fixed. Post-merger prices are thus the fixed point of the system:

$$\tilde{p}_t = \hat{c}_t + \left(\tilde{\mathcal{H}}_t \odot D(\tilde{p}_t, x_t, \hat{\xi}_t; \hat{\theta}^D) \right)^{-1} s(\tilde{p}_t, x_t, \hat{\xi}_t; \hat{\theta}^D). \quad (1)$$

The same procedure applies to alternative assumptions on cost and conduct. For instance, assumptions on cost efficiencies generated by the merger can be easily incorporated by using post-merger cost $\tilde{c}_t$ instead of $\hat{c}_t$ in Equation (1).

When performing market counterfactuals in applications, researchers trade off practicality and data limitations with the dangers of misspecification; in what follows we propose an approach to flexibly estimating supply.

3 Flexible Supply for Market Counterfactuals

3.1 The Flexible Supply Function

Building on the equilibrium framework in Section 2, we develop a flexible approach to modeling supply that maintains economic structure while relaxing parametric assumptions. Under Assumptions 1-4, equilibrium prices satisfy the system of equations $p_{jt} = \Delta_j(s_t, D_t, \mathcal{H}_t) + \bar{c}_j(M_t s_t, w_{jt}) + \omega_{jt}$ for all $j = 1, \dots, J$, where markup functions $\Delta_j(\cdot)$ depend on market shares and demand derivatives, and marginal cost functions $\bar{c}_j(\cdot)$ depend on quantities $q_{jt} = M_t s_{jt}$ and cost shifters. We can thus define:

Definition 1 (*Flexible Supply Function*). For each product $j \in \{1, \dots, J\}$, the *flexible supply function* h_j is the sum of the markup and marginal cost functions:

$$h_j(s_t, D_t, w_{jt}, \mathcal{H}_t, M_t) \equiv \Delta_j(s_t, D_t, \mathcal{H}_t) + \bar{c}_j(M_t s_t, w_{jt}), \quad j = 1, \dots, J.$$

The flexible supply function allows us to express the equilibrium price equation as a nonparametric regression model with an additive, unobservable shock:

$$p_{jt} = h_j(s_t, D_t, w_{jt}, \mathcal{H}_t, M_t) + \omega_{jt}, \quad j = 1, \dots, J. \quad (2)$$

This equation forms the basis of our estimation strategy. The primary restriction is that it does not impose separability between markups and costs; the Online Supplemental Materials (Section S.1) develop extensions that achieve this decomposition. When coupled with a symmetry restriction, our specification efficiently exploits across- and within-market variation present in standard IO data.

In general, the product-specific functions h_j must be estimated separately, one high-dimensional function per product. We reduce this to a single function by imposing exchangeability:

Assumption 6. (*Supply Symmetry*) The supply function satisfies an exchangeability restriction such that for all products $j = 1, \dots, J$:

$$h_j(s_t, D_t, w_{jt}, \mathcal{H}_t, M_t) = h(s_{jt}, s_{-j,t}, D_{jt}, D_{-j,t}, w_{jt}, \mathcal{H}_t, M_t),$$

where $s_{-j,t}$ denotes the vector of rival market shares and D_{jt} and $D_{-j,t}$ partition the demand derivative matrix into own and cross-price derivatives.

Under symmetry, all $J \times T$ product-market observations can be pooled to estimate a single function h , rather than J separate functions. The assumption is satisfied when all firms share the same mode of competition. Under Bertrand, all firms maximize own profits taking rivals' prices as given, so the markup function takes the same form across products up to reordering of arguments. Similarly, Cournot competition and symmetric profit-weight conduct (where all firms place a common weight κ on rivals' profits) also satisfy the restriction. The assumption rules out models with heterogeneous conduct across firms, such as Stackelberg competition, firm-pair-specific collusive weights, or heterogeneous profit weights.⁶

We formally consider formal identification of the flexible supply function next.

3.2 Identification

Identification of the functions h_j in Equation (2) requires addressing the endogeneity of its arguments, (s_t, D_t) , which are simultaneously determined with prices. In turn, identification requires valid instrumental variables. For notational simplicity in the following arguments, we condition on a fixed market size M_t and ownership structure $\mathcal{H}_t$, suppressing them as arguments of the function h_j . The identification results hold conditional on any given values of these market-level variables. Although Assumption 6 restricts attention to a single function h , we state the identification result for the general product-specific functions h_j ; identification of h follows as a special case.

As in the nonparametric identification literature (Newey and Powell, 2003; Blundell, Chen, and Kristensen, 2007; Berry and Haile, 2014), identification is based on the conditional moment restriction:

$$\mathbb{E}[\omega_{jt} \mid z_{jt}, \mathbf{w}_{jt}] = \mathbb{E}[p_{jt} - h_j(s_t, D_t, \mathbf{w}_{jt}) \mid z_{jt}, \mathbf{w}_{jt}] = 0, \quad (3)$$

where z_{jt} denotes instruments for product j in market t , and $\mathbf{w}_{jt}$ represents own exogenous cost shifters that enter the supply function directly.

Two special challenges arise in this setting. First, we must ensure we identify supply rather than inverse demand. For example, under logit demand where $D_t = D(s_t)$, the supply function $h_j(s_t, \mathbf{w}_{jt})$ could potentially replicate inverse demand $p_{jt} =$

⁶Implementing Assumption 6 in settings with varying numbers of firms and products requires data pre-processing to ensure consistent input dimension for estimation; see the Online Supplemental Materials (Section S.2) for a constructive example.

$[\log s_{jt} - \log s_{0t} - x_{jt}^{(1)} - \xi_{jt}]/\alpha_p$. We thus impose exclusion restrictions to prevent this:

Assumption 7 (*Exclusion Restrictions*). There exists a vector of instruments z_{jt} satisfying $\mathbb{E}[\omega_{jt} \mid z_{jt}, w_{jt}] = 0$, where z_{jt} contains the full vector of product characteristics x_{jt} , and $x_{jt}^{(1)}$ is excluded from the cost shifters w_{jt} .

Second, without additional structure, we would need to identify product-specific functions h_j of $J + J^2$ endogenous variables; finding instruments that independently move all the variables (s_t, D_t) as required by standard completeness conditions (Newey and Powell, 2003) would be extremely demanding, if not infeasible in most applications.⁷ Our key insight is that s_t and D_t are linked through the (known) demand function. Thus, each function h_j needs only to be identified on the manifold:

$$\mathcal{M} = \{(s_t, D_t) : D_t = D(s_t, \delta_t, x_t^{(2)}) \text{ for some } (\delta_t, x_t^{(2)}) \in \text{Supp}(\delta_t, x_t^{(2)} \mid s_t)\},$$

where $\delta_t = x_t^{(1)} + \xi_t$ from Assumption 3. This manifold has dimension at most $2J + J(K - 1)$, which is (in typical applications) much lower than the $J + J^2$ dimensions of the full space. We therefore require a completeness condition adapted to the manifold structure:

Assumption 8 (*Manifold Completeness*). For any measurable function $B : \mathcal{M} \times \mathcal{W} \rightarrow \mathbb{R}$ with finite expectation:

$$\mathbb{E}[B(s_t, D_t, w_{jt}) \mid z_{jt}, w_{jt}] = 0 \text{ a.s.} \implies B(s_t, D_t, w_{jt}) = 0 \text{ a.s. on } \mathcal{M}.$$

Under these assumptions, we can prove that the functions h_j are identified:

Theorem 1 (*Identification of Supply*). Under Assumptions 1-5 and 7-8, the supply functions h_j are identified on $\mathcal{M}$ for all $j = 1, \dots, J$.

Proof. See Appendix A. □

Manifold completeness is an adaptation of the completeness conditions in Berry and Haile (2014) to our setting. The key advantage is the dimensional reduction: for $J = 30$ products with $K = 3$ characteristics, the manifold has dimension at most $2J + J(K - 1) = 120$, compared to $J + J^2 = 930$ for the full space, making the

⁷Even exploiting the fact that D_t is symmetric for many commonly used demand systems, we would still need $O(J^2)$ instruments in that case.

completeness requirement substantially weaker. Appendix A provides a detailed discussion, including a comparison with [Berry and Haile \(2014\)](#).

Instrumental Variables: The identification strategy requires instruments that provide two types of variation. With endogenous variables (s_t, D_t) constrained to lie on manifold $\mathcal{M}$, we need excluded instruments only for (s_t, δ_t) . Rival cost shifters $w_{-j,t}$ serve as natural instruments for market shares: when competitor costs change, their equilibrium prices adjust, which in turn affects own market share through substitution patterns. Meanwhile, excluded product characteristics $x_{jt}^{(1)}$ provide variation in demand and its derivatives without directly affecting costs.

The identification strategy can also be illustrated via a practical approach using predicted instruments, similar in spirit to the predicted prices procedures in [Berry et al. \(1999\)](#) and [Gandhi and Houde \(2020b\)](#), though with a key difference. While those papers use predicted prices to identify demand given a known supply model, here we use predicted demand derivatives to identify supply given known demand. Specifically, we can construct predicted market shares $\hat{s}_t$ and predicted demand derivatives $\hat{D}_t$ using the projection of endogenous variables on instruments and the known functional form of demand. These predicted values lie on the manifold $\mathcal{M}$ by construction and can serve as generated instruments in estimation.

Crucially, for each product-specific function h_j , we observe only one realization per market: the equilibrium outcome for product j in market t . Thus, identification of h_j relies entirely on variation in (s_t, D_t, w_{jt}) across markets. With J products and T markets, we effectively have T observations to identify each of the J functions. The symmetry restriction introduced above (Assumption 6) addresses this challenge by pooling all $J \times T$ observations to estimate a single function h .

3.3 Limitations and Comparison with Other Methods

Our framework accommodates several counterfactuals: changes in product characteristics or cost shifters w_{jt} (e.g., product regulation, carbon pricing; [Barwick et al. 2024](#)), adjustments to the tax or subsidy structure (e.g., pass-through analysis, Laffer curves; [Miravete et al. 2018](#)), and modifications to ownership matrices $\mathcal{H}_t$ or product sets $\mathcal{J}$ (e.g., mergers, divestitures; [Miller and Weinberg 2017](#)). These three classes correspond to the counterfactual exercises in Section 5 and Section 6.

The baseline method identifies the supply function h_j as a composite of markups and marginal costs, without separately identifying each component. This suffices for the counterfactuals above, where the object of interest is the equilibrium price response, but precludes measurement of firm profits, total welfare, or counterfactuals that alter the cost or markup function in isolation (e.g., coordinated effects). The Online Supplemental Materials (Section S.1) develop two approaches that achieve this decomposition: one exploiting exogenous market size variation, the other imposing economic restrictions. While the method we propose in this paper is general, specific applications may warrant incorporating additional structure tailored to the institutional setting, and the discussion in Section S.1 shows how these could be implemented.

Applications of counterfactual analysis span a range of empirical environments, and the tradeoffs inherent in our method affect its relative performance; two dimensions are particularly relevant. The first is the exogenous variation available in the data. Our approach, as a nonparametric method, requires the richest identifying variation (Section 3.2). Conduct testing approaches (e.g., Backus et al., 2021; Duarte et al., 2024) require less (Magnolfi and Sullivan, 2022), and parametric approaches that impose a model of competition and cost require none beyond what is needed for demand estimation. The second dimension is extrapolation: how far beyond the training support does the counterfactual lie? Parametric models embed functional form assumptions that allow extrapolation, but at the risk of misspecification. Our nonparametric approach is in principle less reliable far from the training support, though we show in Section 5 that the flexible model maintains good predictive accuracy well beyond it. Related to this observation, the ML literature provides a useful perspective: Balestrieri, Pesenti, and LeCun (2021) show that in high-dimensional settings such as ours, the distinction between interpolation and extrapolation is less sharp than intuition suggests. In addition, our approach imposes no functional form on the supply side, and thus carries the lowest risk of misspecification among the alternatives. How these tradeoffs play out depends on the specific application, which makes our method complementary to existing parametric and testing approaches.

Finally, a natural alternative to our structural approach would be to predict equilibrium prices directly from exogenous variables, bypassing the supply function. However, Berry and Benkard (2006) show that this reduced-form mapping is not nonparametrically identified in differentiated products models: nonseparable unobservables prevent the mapping from exogenous variables to equilibrium prices from being

point-identified. Our approach avoids this problem by conditioning on endogenous equilibrium outcomes. Through the demand inversion (Berry, 1994), observed market shares reveal the realizations of unobserved demand states, allowing us to identify the structural supply function on the relevant manifold. In practice, market shares carry substantial information about the competitive environment that a reduced-form approach does not exploit. The same critique applies to modern ML/AI methods that map exogenous primitives directly to equilibrium prices. A neural network fit on (x_{jt}, w_{jt}) alone remains biased in counterfactuals because of lack of identification, not finite-sample noise. We return to this point as we show simulation results in Section 5.

4 Estimation

While we have shown that our model can be identified using standard sources of variation, estimating high-dimensional nonparametric models in finite samples is notoriously challenging. To this end, we turn our attention to the ML/AI literature and adopt the neural Variational Method of Moments (Bennett and Kallus, 2023), which we refer to as VMM throughout. In this section, we first define the VMM estimator, then discuss the rationale for adopting VMM, and outline the procedures for quantifying uncertainty in estimates.

To introduce VMM, we restart from the moment condition in Equation 3 and define the parameter vector $\theta \in \Theta \subset \mathbb{R}^b$ (where b is potentially very large) that characterizes the flexible supply function h_j :

$$\mathbb{E}[p_{jt} - h_j(s_t, D_t, w_{jt}; \theta) \mid z_t, w_{jt}] = 0.$$

Denote our sample size as $N = TJ$. Given a preliminary consistent estimate $\tilde{\theta}_N$, the VMM estimator $\hat{\theta}_N$ solves a min-max program:⁸

$$\hat{\theta}_N \equiv \operatorname{argmin}_{\theta \in \Theta} \sup_{f \in \mathcal{F}_N} \frac{1}{TJ} \sum_{j,t} f(z_{jt})' \omega_{jt}(\theta) - \frac{1}{4TJ} \sum_{j,t} (f(z_{jt})' \omega_{jt}(\tilde{\theta}_N))^2 - R_N(f, \theta) \quad (4)$$

$$\text{where } \omega_{jt}(\theta) = p_{jt} - h_j(s_t, D_t, w_{jt}; \theta).$$

⁸Recent ML theory literature (e.g., Zhu, Zhang, Wang, Yang, and Chen, 2024; Chen, Chen, Qi, Chen, and Yang, 2025b) advances the frontier by studying min-max-based estimation problems of this type in nonparametric IV settings.

For our implementation, $f \in \mathcal{F}_N$ and $h_j(\cdot; \theta) \in \mathcal{H}_N$ are classes of neural networks with growing width and depth; $R_N : \mathcal{F}_N \times \mathcal{H}_N \rightarrow [0, \infty]$ penalizes their complexity. Regularization can also enforce economic properties of the supply function (Online Supplemental Materials, Section S.1.3). Under regularity conditions on the parameter space and function classes, including compactness (natural as prices and markups are bounded), and standard conditions on $\mathcal{F}_N$ and $\mathcal{H}_N$ (satisfied by neural network architectures with growing width and depth), Theorem 4 of [Bennett and Kallus \(2023\)](#) establishes consistency of $\hat{\theta}_N$ for the true θ_0 .

4.1 Discussion of VMM

The VMM estimator bears a close relationship to GMM, the workhorse estimator in structural econometrics. In GMM, the researcher selects a fixed set of moment functions and constructs the optimal weighting matrix. The quality of the estimator depends on how well the moments detect violations of the identifying restrictions. The optimal instrument ([Chamberlain, 1987](#)) would maximize efficiency, but computing it is generally unfeasible. This is a familiar problem for IO researchers, e.g., in the demand estimation literature following [Berry et al. \(1995\)](#), where considerable effort has been devoted to constructing approximations to optimal instruments. VMM automates this search. The “adversary” network $f(z)$ learns the function that most effectively detects moment violations for the current supply estimate, while the supply network h adjusts to eliminate them. The supremum over f in Equation (4) searches for the most informative moment condition; the quadratic penalty term assigns optimal weights using the preliminary estimate $\tilde{\theta}_N$. When $\mathcal{F}_N$ is a finite-dimensional span $\{f_1, \dots, f_k\}$, VMM coincides with optimally weighted GMM (OWGMM).

To make this equivalence precise, we reproduce the proof of Lemma 1 from [Bennett and Kallus \(2023\)](#). Let $\tilde{\theta}_N$ denote a preliminary estimate and Γ_N the optimal weighting matrix based on $\tilde{\theta}_N$. Then, $\hat{\theta}^{\text{OWGMM}}(f_1, \dots, f_k, \tilde{\theta}_N) = \hat{\theta}_N^{\text{VMM}}(\text{span}\{f_1, \dots, f_k\}, \tilde{\theta}_N)$. This is because, by definition of OWGMM,

$$\begin{aligned} \hat{\theta}_N^{\text{OWGMM}}(f_1, \dots, f_k, \tilde{\theta}_N) &= \arg \min_{\theta \in \Theta} \left\| \Gamma_N^{-1/2} \mathbb{E}_N[f(z)\omega(\theta)] \right\|^2 \\ &= \arg \min_{\theta \in \Theta} \sup_{v \in \mathbb{R}^k} \left\{ v' \mathbb{E}_N[f(z)\omega(\theta)] - \frac{1}{4} v' \Gamma_N v \right\} \\ &= \arg \min_{\theta \in \Theta} \sup_{v \in \mathbb{R}^k} \left\{ \mathbb{E}_N[(f(z)'v)' \omega(\theta)] - \frac{1}{4} \mathbb{E}_N[((f(z)'v)' \omega(\tilde{\theta}_N))^2] \right\}. \end{aligned}$$

The first equality follows from the dual norm representation, and the second uses the fact that $f(z)'v \in \text{span}\{f_1, \dots, f_k\}$. When $\mathcal{F}_N$ is a rich class of neural networks, the adversary searches over a large space of test functions, effectively learning nonlinear combinations of instruments that are most informative for identifying supply.

In principle, our supply function can be estimated with standard nonparametric instrumental variable (NPIV) estimation methods (see the reviews of, e.g., [Chen, 2007](#); [Carrasco, Florens, and Renault, 2007](#)); these methods are well understood, and provide flexibility in modeling with robust asymptotic properties. However, a key advantage of the neural VMM implementation is its ability to cope with the curse of dimensionality. To fix ideas, consider our setting with $J = 30$ products. The supply function h depends on 30 market shares, a 30×30 matrix of demand derivatives, and cost shifters, yielding over 930 arguments. For standard sieve-based NPIV, the minimax convergence rate scales as $n^{-2s/(2s+d)}$, where d is the ambient dimension and s the smoothness order; with $d \approx 930$, this rate is prohibitively slow.

Deep neural networks circumvent this problem by exploiting compositional structure in the target function ([Bauer and Kohler, 2019](#); [Schmidt-Hieber, 2020](#)). Supply functions derived from oligopoly models are inherently compositional. Bertrand markups, for instance, can be decomposed as $\Delta_j = e_j'(\mathcal{H}_t \odot D_t)^{-1}s_t$, where $\mathcal{H}_t$ is the ownership matrix and D_t the matrix of demand derivatives. The ownership matrix has a block-diagonal structure, with blocks corresponding to each firm's product set. The Hadamard product $\mathcal{H}_t \odot D_t$ zeros out cross-firm entries, so the inversion decomposes into independent blocks of dimension at most $J_f \times J_f$, where J_f is the largest firm's product count. Each intermediate computation thus depends on at most $O(J_f^2)$ variables rather than $O(J^2)$. [Schmidt-Hieber \(2020\)](#) show that when a target function admits such compositional structure, neural networks achieve approximation rates depending on the intrinsic dimensions of each component rather than the ambient dimension d . For Bertrand markups with, e.g., 5 firms of 6 products each, the binding intrinsic dimension is $J_f^2 = 36$, the size of each firm's demand derivative block, compared to the ambient $d = 930$ (see [Appendix B](#) for a formal treatment). While we do not know for certain the true nature of conduct in the data, we note that many standard models, including profit-weight and Cournot, exhibit the same block structure and thus the same favorable approximation properties.

A potential shortcoming of VMM is that the minimax formulation in [Equation \(4\)](#) involves a saddle-point optimization, which is more demanding than standard loss

minimization: the stability of such procedures in finite samples is a legitimate concern. In our implementation, we find good finite-sample performance in the Monte Carlo evidence of Section 5, perhaps helped by regularization through early stopping on a validation set, and by the compositional structure of the target function.

4.2 Quantification of Uncertainty

We now turn to the quantification of uncertainty in counterfactual predictions. Although the supply function is nonparametrically identified and estimated, following [Bennett and Kallus \(2023\)](#), uncertainty is quantified by imposing a flexible parametric structure based on a neural network architecture, whose parameters lie in a compact subset of a finite-dimensional Euclidean space. The general principle that parametric inference can remain valid within sufficiently flexible parametric families has support in the semiparametric literature (e.g., [Ai and Chen, 2007](#); [Ackerberg, Chen, and Hahn, 2012](#)). Fully nonparametric inference lies beyond the scope of this paper and is left for future research.⁹ For the asymptotic analysis, we consider regimes in which J remains fixed while $N \rightarrow \infty$ (equivalently, $T \rightarrow \infty$).

While [Bennett and Kallus \(2023\)](#) provides valid element-wise testing procedures for the neural network parameters, our focus is on counterfactual prices, which are functionals of these parameters. A merger simulation, for instance, predicts a new equilibrium price for every product in every affected market; a policy evaluation requires confidence intervals for the price of each affected product. This differs from standard parametric settings, where inference targets a low-dimensional parameter vector. Here, we require *simultaneous* confidence intervals across $\bar{d}$ product-market observations, which we construct by combining the numerical delta method with Holm’s step-down procedure ([Holm, 1979](#)) and a permutation algorithm.

We now describe the procedure; formal statements are in Appendix C. For notational convenience, we suppress the dependence on observables and index the $\bar{d}$ product-market observations at which inference is conducted by $\ell = 1, \dots, \bar{d}$; each ℓ corresponds to a specific product j in market t , so that $h^{(\ell)}(\theta) \equiv h_j(s_t, D_t, w_{jt}; \theta)$ and the vector $h(\theta) = (h^{(1)}(\theta), \dots, h^{(\bar{d})}(\theta))' \in \mathbb{R}^{\bar{d}}$ collects predicted prices. Suppose that a

⁹Our inference procedures condition on first-step demand estimates. Standard two-step inference frameworks (e.g., [Newey and McFadden, 1994](#)) provide corrections that apply to our case, and under sufficient rate conditions on the first-step demand estimator (e.g., if $\hat{D}_t$ converges at a $\sqrt{T}$ -rate), the impact on second-step supply estimation is asymptotically negligible (cf. [Farrell, Liang, and Misra, 2020](#)). Studying joint nonparametric estimation of demand and supply is a direction for future work.

consistent first-stage estimator $\tilde{\theta}_N$ is available. Under standard regularity conditions (Equation (C1) in Appendix C):

$$\{\nabla_{\theta'} h(\theta_0) \Omega_0^{-1} \nabla_{\theta'} h(\theta_0)' / N\}^{-1/2} (h(\hat{\theta}_N) - h(\theta_0)) \xrightarrow{d} N(0, I).$$

The asymptotic variance is generally unknown. The Jacobian $\nabla_{\theta} h(\theta_0)$ has dimension $\bar{d} \times b$. For a single observation ($\bar{d} = 1$), Lemma 9 in [Bennett and Kallus \(2023\)](#) provides a tractable characterization: for any $\lambda \in \mathbb{R}^b$,

$$\lambda^T \Omega_0^{-1} \lambda = -\frac{1}{4} \inf_{\gamma \in \mathbb{R}^b} \sup_{f \in \mathcal{F}} \mathbb{E}[f(z)' \nabla_{\theta} \omega(\theta_0) \gamma] - \frac{1}{4} \mathbb{E}[(f(z)' \omega(\theta_0))^2] - 4\gamma' \lambda - R_N(f). \quad (5)$$

Taking $\lambda = \nabla_{\theta} h^{(\ell)}(\theta_0)$, the solution yields the asymptotic variance for observation ℓ . The gradient $\nabla_{\theta} h^{(\ell)}(\theta_0)$ is computed via automatic differentiation,¹⁰ and $\hat{\theta}_N$ replaces θ_0 in practice.

When $\bar{d} \geq 2$, this approach does not directly yield a joint covariance matrix. We construct simultaneous confidence intervals by adapting Holm’s step-down procedure with the variance estimates $\hat{\sigma}_{\ell}^2(\hat{\theta})$ and predictions $h^{(\ell)}(\hat{\theta})$ for each $\ell = 1, \dots, \bar{d}$. The critical values T_{α_k} correspond to adjusted significance levels $\alpha_k = \alpha / (\bar{d} + 1 - k)$ for $k = 1, \dots, \bar{d}$. Because the ordering of p -values is unknown, we construct intervals for all $\bar{d}!$ permutations and take their union. The procedure is summarized in Algorithm 1.

Algorithm 1 Simultaneous Confidence Interval for neural VMM

- 1: **for** each product-market observation $\ell \in \{1, \dots, \bar{d}\}$ **do**
 - 2: Estimate $\hat{\sigma}_{\ell}^2(\hat{\theta})$ by solving Equation 5 with $\lambda = \nabla_{\theta} h^{(\ell)}(\hat{\theta})$
 - 3: **end for**
 - 4: Fix critical values $T_{\alpha} = \{T_{\alpha_k} : k = 1, \dots, \bar{d}\}$ where $\alpha_k = \frac{\alpha}{\bar{d} + 1 - k}$
 - 5: **for** each permutation $\tilde{\ell}_1, \dots, \tilde{\ell}_{\bar{d}}$ of $\{1, \dots, \bar{d}\}$ **do**
 - 6: Construct bounds as $h^{(\tilde{\ell}_k)}(\hat{\theta}) \pm N^{-\frac{1}{2}} \hat{\sigma}_{\tilde{\ell}_k}(\hat{\theta}) T_{\alpha_k}$ for $k = 1, \dots, \bar{d}$
 - 7: **end for**
 - 8: Return simultaneous confidence interval as union of bounds across permutations
-

Beyond prices, applied counterfactual analysis often requires inference on quantities such as consumer surplus, market shares, or tax revenue. These are smooth

¹⁰We use `torch` automatic differentiation. The validity of the numerical delta method requires $\epsilon \rightarrow 0$ and $N\epsilon / \log N \rightarrow \infty$ ([Hong, Mahajan, and Nekipelov, 2015](#), Theorem 1).

functionals of the counterfactual price vector: writing $\tilde{p} = h(\theta)$ for the vector of counterfactual prices, a quantity of interest takes the form $F(\tilde{p})$ for some differentiable $F : \mathbb{R}^{\tilde{d}} \rightarrow \mathbb{R}$. The delta method extends the result above:

$$\left\{ \nabla_p F(\tilde{p}_0) \nabla_\theta h(\theta_0) \Omega_0^{-1} \nabla_\theta h(\theta_0)' \nabla_p F(\tilde{p}_0)' / N \right\}^{-1/2} (F(h(\hat{\theta}_N)) - F(h(\theta_0))) \xrightarrow{d} N(0, I),$$

where $\tilde{p}_0 = h(\theta_0)$ and $\nabla_p F$ denotes the gradient of F with respect to prices. Formal derivations are provided in Appendix C. We illustrate with an example relevant to merger analysis and policy evaluation.

Example 3 (Counterfactual welfare and revenue). *Market shares. For the counterfactual share of product j , take $F(\tilde{p}) = s_j(\tilde{p})$, so that $\nabla_p F = D_j(\tilde{p})$, the j -th row of the demand Jacobian evaluated at counterfactual prices.*

Consumer surplus. *Under logit demand across markets $m = 1, \dots, M$:*

$$F(\tilde{p}) = -\frac{1}{\alpha_p} \sum_m \log \left(1 + \sum_{j \in \mathcal{I}_m} \exp(-\alpha_p \tilde{p}_j + x_j \beta + \xi_j) \right),$$

and $\nabla_p F = s(\tilde{p})$, the vector of market shares at counterfactual prices.

Tax revenue. *For revenue from ad valorem taxes with tax-inclusive prices, let $a = (a_j)_{j \in \mathcal{G}}$ denote the tax share of consumer expenditure. Then $F(\tilde{p}) = \sum_{j \in \mathcal{G}} a_j \tilde{p}_j s_j(\tilde{p})$ with gradient $\nabla_p F = a \odot s(\tilde{p}) + (a \odot \tilde{p})' D(\tilde{p})$, where $D(\cdot)$ is block-diagonal across markets.*

5 Monte Carlo Simulations

We evaluate five dimensions of performance through Monte Carlo simulations: (i) predictive accuracy in hold-out samples across sample sizes from $T = 100$ to $T = 10,000$; (ii) scalability to high-dimensional environments with 30 products and over 900 supply function arguments; (iii) economic interpretability through pass-through analysis; (iv) counterfactual prediction for product regulations, tax policies, and mergers, including extrapolation beyond training support; and (v) reliability of our inference procedure in finite samples.

5.1 Simulation Environments

We develop three environments of increasing complexity. In each, we generate data under four supply-side specifications at multiple sample sizes, crossing two conduct models (Bertrand and a profit-weight model with weight κ on rivals' profits) with two cost structures (constant or decreasing marginal cost). This 2×2 crossing yields four distinct data-generating processes per environment. Appendix D fully describes these environments and all simulation details.¹¹ All simulation environments satisfy Assumption 6: the conduct models we consider all have symmetric supply functions.

Baseline Environment. Markets feature 2 or 3 single-product firms. On the demand side, we adopt a logit specification. Demand is known to the researcher (we use the true demand derivatives), so the simulation results isolate the performance of the supply-side estimator without confounding from first-step estimation error. When considering profit-weight models, we set $\kappa = 0.5$. We generate samples with $T \in \{100, 1,000, 10,000\}$ markets.

High-Dimensional Environment. To address concerns about the curse of dimensionality, we implement a second environment, inspired by the setting of Miller and Weinberg (2017), featuring 30 differentiated products per market – an order of magnitude larger than our baseline. This yields a $30 \times 30 = 900$ dimensional demand derivative matrix. We adopt a more flexible nested logit demand system than in our baseline environment and generate samples with $T \in \{100, 1,000, 10,000\}$. When considering profit-weight models, we set $\kappa = 0.75$.

Merger Simulation Environment. For merger counterfactuals (Section 5.5), we augment the high-dimensional environment with richer ownership variation. Markets contain either three firms (with 6, 5, and 4 products) or four firms (one each with 5 and 4 products, two with 3 products), with 0–5 products randomly dropped per market to yield 10–15 active products. This ensures that the pre-merger data contains variation in ownership ($\mathcal{H}_t$) analogous to the merger we simulate, helping the flexible model learn how the change in market structure will affect equilibrium pricing. When considering profit-weight models, we set $\kappa = 0.75$. We use $T = 1,000$ markets.

¹¹The Supplemental Materials (Section S.3) report additional simulation results.

Crucially for identification, markets in all data environments feature variation in both observed cost shifters w_{jt} and product characteristics x_{jt} excluded from cost, allowing us to form the necessary instrumental variables as discussed in Section 3.2. Our simulation strategy is summarized in Appendix Table D1, which lists the specific data environments we use when evaluating each dimension of our model’s performance.

Throughout this section, brackets around estimates report the 10th and 90th percentiles across 50 simulation runs; for the hold-out exercises, we re-initialize and refit the model in each run but keep ξ_{jt} and ω_{jt} constant, while for the counterfactual exercises each run also redraws the unobservables. Thus, these ranges reflect variability from model fit (and from the simulated data in counterfactuals). We reserve the term *confidence interval* for Section 5.6, which implements the inference procedure of Section 4.

5.2 Hold-out Sample Performance

We examine the ability of $\hat{h}$ to predict prices in hold-out markets not used during estimation, in our baseline environment with 2 or 3 products per market. We randomly sample 80% of markets for training and reserve the remaining 20% as a hold-out test sample.¹² For each hold-out market, we predict prices using the estimated flexible supply function and compare these to predictions from standard parametric models. We consider three parametric benchmarks: Bertrand, joint profit maximization (monopoly), and perfect competition, all with constant marginal cost.

The implementation details for both the flexible and parametric models, including choice and formation of instruments and computation of counterfactuals, are provided in Appendix D. For the flexible model, the primary implementation decision involves choosing the neural network architecture. We experiment with three hidden layer widths $|h|$: a “small” 3×3 ($|h| = 3$), a “medium” 20×20 ($|h| = 20$), and a “large” 100×100 ($|h| = 100$) specification. We also report results from a flexible model that omits demand derivatives D_t as an argument to assess their importance.

Finally, we implement the reduced-form approach discussed in Section 3.3 with an ML/AI approach: we train a transformer (Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, and Polosukhin, 2017; Lee, Lee, Kim, Kosior, Choi, and Teh,

¹²Within the training sample, 20% is reserved as a hold-out validation set used for early stopping.

2019) on (x_{jt}, w_{jt}) alone, without equilibrium shares as inputs.¹³

TABLE 1: MSE Ratios Across Models, Baseline Environment with Economies of Scale

T	Standard Models			Flexible Models			TF-RF	D_t
	B	M	P	$ h = 3$	$ h = 20$	$ h = 100$		
Panel A: Bertrand-Nash DGP								
100	2.28	552.52	7.04	[1.27, 1.35]	[1.23, 1.31]	[0.99, 1.08]	[5.92, 9.78]	No
				[1.13, 1.20]	[1.14, 1.20]	[1.03, 1.10]		Yes
1,000	2.73	625.68	9.73	[1.33, 1.34]	[1.22, 1.34]	[1.20, 1.27]	[5.67, 6.78]	No
				[1.40, 1.43]	[1.40, 1.42]	[1.16, 1.24]		Yes
10,000	2.95	933.55	9.61	[1.44, 1.45]	[1.24, 1.43]	[1.03, 1.09]	[5.65, 6.10]	No
				[1.41, 1.43]	[1.34, 1.43]	[1.00, 1.07]		Yes
Panel B: Profit-Weight DGP								
100	4.39	41.43	7.91	[1.60, 1.67]	[1.61, 1.67]	[1.55, 1.81]	[5.87, 11.38]	No
				[1.31, 1.36]	[1.33, 1.39]	[1.37, 1.49]		Yes
1,000	6.07	33.57	9.79	[1.99, 2.01]	[1.15, 1.65]	[1.08, 1.16]	[5.94, 7.36]	No
				[1.48, 1.54]	[1.07, 1.49]	[1.02, 1.07]		Yes
10,000	6.45	46.78	9.81	[2.14, 2.15]	[1.07, 1.38]	[1.03, 1.06]	[5.87, 6.29]	No
				[1.61, 1.63]	[1.05, 1.59]	[1.00, 1.01]		Yes

Notes: MSE ratios relative to the correctly specified model (ratio of 1 = equal performance). B = Bertrand, M = monopoly, P = perfect competition, all with constant cost. TF-RF = reduced-form transformer trained on cost shifters and product characteristics. MSE computed on a hold-out test sample of markets not used in training. Brackets are the 10th and 90th percentiles across 50 simulation runs.

Table 1 reports mean squared error (MSE) ratios for price predictions, expressed relative to the correctly specified parametric model.¹⁴ A ratio of 1 indicates performance equal to the true model; values above 1 indicate worse performance; the correctly specified model’s MSE is the irreducible error from cost shocks.¹⁵ Data are generated under Bertrand conduct (Panel A) and profit-weight conduct (Panel B), both with economies of scale. The latter cost structure generates an additional challenge as it requires $\hat{h}$ to learn how both markups and marginal costs vary with quantities.

The flexible model achieves MSEs close to the correctly specified model while significantly outperforming misspecified parametric alternatives and the reduced form

¹³See Appendix B.1 for architecture details.

¹⁴The mean squared error for price predictions is computed as $\text{MSE} = \frac{1}{N_h} \sum_{t \in \mathcal{T}_h} \sum_{j=1}^{J_t} (\hat{p}_{jt} - p_{jt})^2$, where $\hat{p}_{jt}$ denotes the predicted price and p_{jt} the true price for product j in market t , $\mathcal{T}_h$ is the hold-out test sample, J_t is the number of products in market t , and $N_h = \sum_{t \in \mathcal{T}_h} J_t$ is the total number of product-market observations in the hold-out sample.

¹⁵Results for constant marginal cost DGPs are similar and reported in Appendix Table S.3.5.

approach. In Panel B (profit-weight conduct with economies of scale), the flexible model with $T = 10,000$ and a large network achieves an MSE ratio that nearly matches the true model, while the misspecified Bertrand model’s ratio is 6.45, over six times larger.

Even with just $T = 100$ markets, the flexible model outperforms misspecified parametric alternatives – in fact, improvements with sample size are moderate in this simple environment. As this baseline features logit demand, including demand derivatives D_t provides only modest improvements, consistent with the simple substitution patterns.¹⁶ Similarly, while larger networks ($|h| = 100$) perform somewhat better than smaller ones, even the small 3×3 network substantially outperforms misspecified parametric models. This suggests that in simpler market settings, model flexibility matters more than network capacity or derivative information.

Table 1 also includes the reduced-form transformer (TF-RF column), which plateaus at MSE ratios between 5.7 and 6.3 at $T = 10,000$ across DGPs. This gap to our flexible model, which includes information from endogenous outcomes and instruments, reflects the identification failure discussed in Section 3.3, not finite-sample noise.

5.3 Scalability to High-Dimensional Environments

A fundamental limitation of nonparametric methods is the curse of dimensionality. In our application, the input dimensionality of the supply function grows quadratically with the number of products J because it includes both market shares $s_t \in \mathbb{R}^J$ and demand derivatives $D_t \in \mathbb{R}^{J \times J}$. To test whether our flexible model estimated with VMM can overcome this challenge, we perform simulations in an environment mimicking Miller and Weinberg (2017). With a richer demand system and $J = 30$ products (an order of magnitude larger than in our baseline setting), there are over 900 demand derivative arguments. These derivatives take a more complex form than those in our baseline environment.

Table 2 presents results from this high-dimensional environment. In this more complex setting, the importance of model flexibility and of providing derivatives becomes apparent. Under the Bertrand DGP (Panel A), the flexible model with $T = 1,000$ and demand derivatives achieves MSE ratios close to 1, while substantially outperforming misspecified models.

¹⁶Under logit demand, demand derivatives are a simple function of pairs of market shares. Thus, the neural network can learn D_t from just s_t , and the input D_t becomes redundant.

TABLE 2: MSE Ratios Across Models, High-dimensional Environment with Constant Costs

T	Standard Models			Flexible Models			TF-RF	D_t
	B	M	P	$ h = 3$	$ h = 20$	$ h = 100$		
Panel A: Bertrand-Nash DGP								
100	1.00	4.49	2.49	1.21	1.22	1.23	[2.75, 5.49]	No
				1.22	1.23	1.25		Yes
1,000	1.00	5.87	3.32	1.27	1.08	1.05	[3.36, 4.15]	No
				1.23	1.19	1.20		Yes
Panel B: Profit-Weight DGP								
100	2.43	2.97	2.63	2.62	2.75	2.36	[3.43, 9.16]	No
				2.20	2.32	2.30		Yes
1,000	4.76	5.28	5.32	4.83	3.82	1.50	[4.39, 6.06]	No
				1.56	1.60	1.46		Yes

Notes: MSE ratios relative to the correctly specified model (ratio of 1 = equal performance). B = Bertrand, M = monopoly, P = perfect competition, all with constant cost. TF-RF = reduced-form transformer trained on cost shifters and product characteristics. MSE computed on a hold-out test sample of markets not used in training.

The value of demand derivatives is particularly striking under the profit-weight DGP (Panel B), where strategic complementarities interact with complex substitution patterns. With $T = 1,000$, including derivatives reduces the MSE ratio from 4.83 to 1.56 for the small network and from 3.82 to 1.60 for the medium network. Larger networks also become important: with derivatives, the large network achieves a ratio of 1.46 compared to 1.56–1.60 for smaller networks. As market complexity increases, both network capacity and derivative information become essential for capturing equilibrium relationships; the reduced-form transformer (TF-RF column), which lacks access to either shares or derivatives, plateaus at MSE ratios of 4.4 to 6.1.

These results demonstrate that VMM handles high-dimensional problems well: the network learns which of the more than 900 potential interactions matter for pricing, guided by the moment conditions. A notable feature of this setting is that the network operates in the overparameterized regime: the large network has over 100,000 parameters estimated from only $T = 1,000$ markets, and with $T = 100$ the ratio of parameters to markets exceeds 1,000. This is consistent with recent findings on double descent, whereby test performance improves once the model’s complexity well exceeds the sample size, as gradient-based optimization implicitly selects low-complexity solutions (e.g., [Belkin, Hsu, Ma, and Mandal, 2019](#)).

5.4 Interpretation via Pass-Through Analysis

A common criticism of ML/AI methods in economics is their “black box” nature. To address this, we examine whether $\hat{h}$ learns economically meaningful supply-side relationships by analyzing implied pass-through matrices.

Pass-through matrices capture how cost shocks affect equilibrium prices, reflecting both direct effects of own-cost changes and strategic responses to rivals’ price adjustments. We compute pass-through by increasing costs by 1 percent, solving for new equilibrium prices, and calculating the corresponding percentage price changes.

TABLE 3: Simulated Pass-through Matrices

Panel A: Bertrand DGP				Panel B: Profit-Weight DGP			
True Model		Flex. Model		True Model		Flex. Model	
0.41	0.03	0.47	0.08	0.38	−0.28	0.56	−0.24
		[0.39, 0.68]	[0.03, 0.15]			[0.35, 0.84]	[−0.28, −0.03]
0.13	0.91	0.08	0.72	0.03	0.95	0.00	0.91
		[0.04, 0.15]	[0.50, 0.91]			[−0.18, 0.00]	[0.58, 0.97]

Notes: Simulated pass-through matrices for a representative duopoly market. Element P_{ij} : percentage change in p_{it} from a 1% cost shock to product j . Flexible model trained on $T = 1,000$ markets. For the flexible model, each cell reports the median and (in brackets) the 10th and 90th percentiles across 50 simulation runs.

Table 3 presents pass-through matrices for a representative duopoly market under different data-generating processes in our baseline environment.¹⁷ We train the flexible model using $T = 1,000$ markets, using a medium size network ($|h| = 20$), and including as an argument the matrix D_t . Panel A shows results under Bertrand competition with constant marginal cost, while Panel B presents results under profit-weight conduct. The flexible model’s implied pass-through matrices closely match the truth in both cases.

Under Bertrand competition (Panel A), the flexible model estimates own-cost pass-through of 0.47 and 0.72 on the diagonal, compared to 0.41 and 0.91 for the true model. Cross-cost pass-through values are both 0.08 for the flexible model, versus 0.03 and 0.13 for the true model. While not exact, these estimates correctly capture the key economic features: substantial own-cost pass-through and positive but smaller cross-cost effects reflecting strategic complementarities.

¹⁷These results are for the median duopoly market by inside share; results are robust across markets.

Under profit-weight conduct (Panel B), our estimated flexible model implies own-cost pass-through of 0.56 and 0.91 (versus true values of 0.38 and 0.95). Notably, the flexible model correctly identifies negative cross-cost pass-through (-0.24 versus true value of -0.28), a distinctive feature of partial collusion where firms internalize effects on rivals’ profits. This sign reversal from the Bertrand case demonstrates that the flexible model can learn different modes of conduct without any direct assumptions on the game-theoretic structure.

5.5 Market Counterfactuals

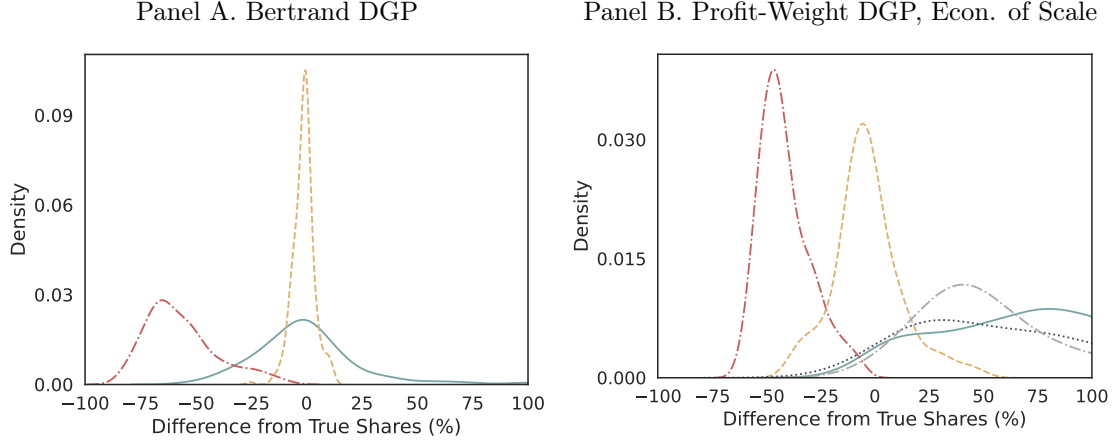
We now turn to simulations that showcase the core application of our method: predicting market outcomes under counterfactual scenarios. Following the framework developed in Section 2, we examine three classes of counterfactuals that span the range of policy interventions commonly studied in empirical IO: (i) product regulation through modification of characteristics, (ii) tax policy design via Laffer curve analysis, and (iii) merger simulation through changes in ownership structure.¹⁸ For each counterfactual, we solve for new equilibrium prices under both the flexible model and various parametric specifications, obtain counterfactual objects under different transformations F , and then compare predictions to the true counterfactuals. Throughout, we focus on one key issue: how far out of the support of the data can our flexible, data-driven model give useful counterfactual predictions?

Product Characteristic Regulation: Governments frequently regulate product characteristics to influence consumer behavior, from fuel economy and air pollution standards (e.g., Ito and Sallee, 2018; Jacobsen, Sallee, Shapiro, and Van Benthem, 2023) to sugar content limits. We examine counterfactuals where, in our baseline environment, characteristic x_1 is shifted to $\tilde{x}_1 = x_1 + 1$. This pushes the counterfactual well beyond the training support where $x_1 \sim U[0, 1]$, thus testing the flexible model’s ability to predict equilibrium responses to product changes not observed in the data.

Figure 1 shows that the flexible model accurately predicts consumption (market share) changes due to regulations on product characteristics, achieving small absolute error across DGPs and beating misspecified parametric models. For instance,

¹⁸The Online Supplemental Materials (Section S.3) report additional results, including robustness exercises, other counterfactuals (changes in product characteristics that affect cost and product exit), and inference.

FIGURE 1: Regulation of Product Characteristics: Share Predictions



Fitted Model	DGP			
	Bertrand	Profit-Weight	Bertrand (Scale)	Profit-Weight (Scale)
--- Bertrand (Scale)	-	-	-	[77.45, 87.40]
..... Bertrand (Const.)	-	[54.04, 56.41]	[12.92, 14.22]	[97.56, 106.26]
--- Monopoly	[58.37, 58.86]	[51.09, 51.36]	[58.28, 59.08]	[42.89, 44.15]
— Perf. Comp.	[23.31, 24.78]	[60.64, 61.96]	[25.35, 26.62]	[77.46, 79.97]
--- Flex Supply	[3.70, 5.28]	[9.01, 11.64]	[9.53, 11.43]	[17.15, 23.57]
TF-RF	[42.57, 48.77]	[38.43, 43.94]	[41.47, 47.14]	[37.23, 42.72]

Notes: Share prediction errors when $\tilde{x}_1 = x_1 + 1$ shifts out of training support. Flexible model: $|h| = 20$, D_t included, $T = 1,000$. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plots show the flexible model with the median mean-squared error across the 50 runs.

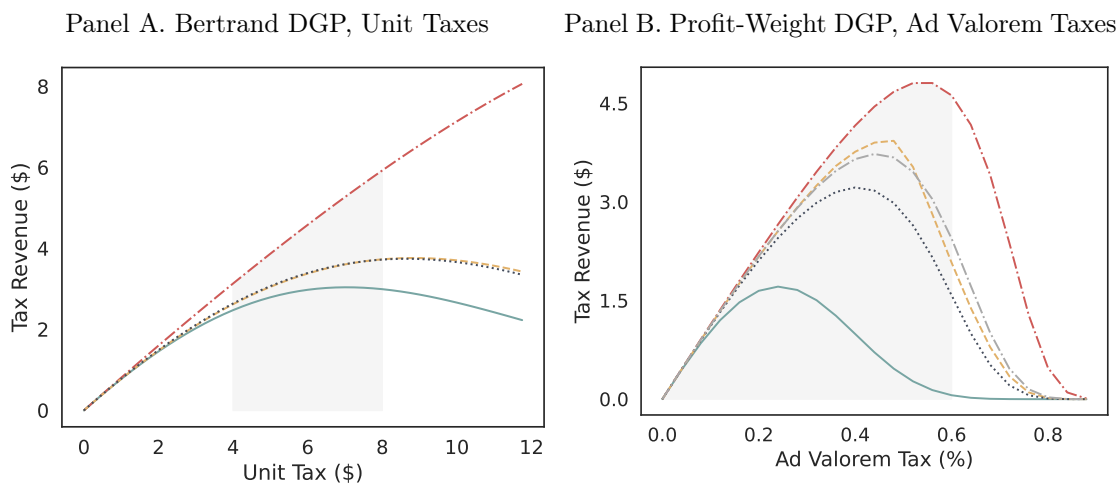
under the profit-weight DGP with economies of scale, the flexible model achieves a root mean percentage squared error (RMPSE) between 17.15% and 23.57%,¹⁹ while all parametric models achieve RMPSEs above 40%. Moreover, the flexible model's distribution of prediction errors is tightly centered around zero, while misspecified models show systematic biases. Figure 1 also reports the TF-RF row, with RMPSEs between 37% and 49% across DGPs.

Tax Policy (Laffer Curves): Tax policy design requires understanding the Laffer curve, or how revenues vary with tax rates (e.g., [Miravete et al., 2018](#)). This is a

¹⁹This is defined as $\text{RMPSE} = 100 \times \sqrt{\frac{1}{N_c} \sum_{t,j} \left(\frac{\tilde{s}_{jt}^m - \tilde{s}_{jt}}{\tilde{s}_{jt}} \right)^2}$, where $\tilde{s}_{jt}^m$ denotes the predicted counterfactual share under model m , $\tilde{s}_{jt}$ is the true counterfactual share, and N_c is the total number of product-market observations in the counterfactual.

highly nonlinear relationship that depends critically on firm conduct and cost structure. We examine both unit taxes τ (levied on firms) and ad valorem tax rates v (levied on consumers), as formalized in Equation (D1) in Appendix D. Training data include tax variation with $v_t \sim U[0, 0.6]$ and $\tau_t \sim U[4, 8]$, but we evaluate predictions at rates up to 90% for ad valorem and \$12 for unit taxes, well outside the training support.²⁰

FIGURE 2: Laffer Curves for Unit and Ad Valorem Taxes



Panel C. MSE

Fitted Model	DGP			
	Bertrand (U)	Profit-Weight (U)	Bertrand (AV)	Profit-Weight (AV)
..... Bertrand	-	[0.21, 0.37]	-	[0.36, 0.65]
--- Monopoly	[1.03, 1.85]	[0.42, 0.79]	[2.09, 3.28]	[0.87, 2.10]
— Perf. Comp.	[0.34, 0.71]	[0.56, 0.98]	[1.03, 1.73]	[1.19, 1.92]
- - - Flex Supply	[0.01, 0.04]	[0.02, 0.03]	[0.03, 0.08]	[0.06, 0.14]
TF-RF	[1.19, 3.29]	[1.07, 3.01]	[1.70, 7.09]	[1.50, 7.26]

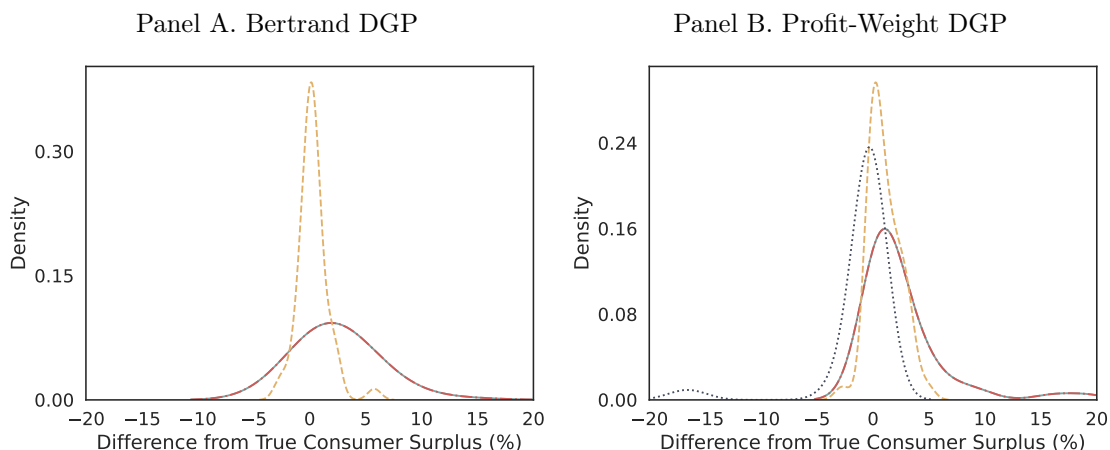
Notes: Predicted Laffer curves under different DGPs and fitted models. Flexible model: $|h| = 20$, D_t included, $T = 1,000$. Shaded regions indicate training support of tax rates. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

Figure 2 demonstrates the flexible model’s ability to capture the nonlinear relationship between tax rates and revenues. In Panel A (Bertrand DGP, unit taxes), the flexible model’s Laffer curve virtually overlays the true curve, correctly identifying

²⁰In Appendix Figure S.3.6, we reduce the variation in tax rates found in the training data. The results show that variation in ad valorem taxes is especially useful for learning the Laffer curve.

the revenue-maximizing rate around \$9. The misspecified perfect competition model substantially underestimates revenues at all rates, predicting a revenue-maximizing rate around \$6. For ad valorem taxes under profit-weight conduct (Panel B), the flexible model achieves an MSE between 0.06 and 0.14 in revenue predictions,²¹ compared to 0.36–0.65 for Bertrand and 0.87–2.10 for monopoly. Figure 2 also reports the TF-RF row, with revenue MSEs between 1.1 and 7.3 across DGPs and tax types.

FIGURE 3: Merger Simulation: Consumer Surplus Changes



Panel C. RMPSE

Fitted Model	Panel DGP	
	Bertrand	Profit-Weight
..... Bertrand (Const.)	-	[3.26, 3.97]
--- Monopoly	[7.90, 10.12]	[4.55, 5.21]
— Perf. Comp.	[7.90, 10.12]	[4.55, 5.21]
-.- Flex Supply	[1.44, 1.94]	[1.53, 1.94]
TF-RF	[7.12, 12.11]	[17.65, 25.80]

Notes: Distribution of percentage differences between predicted and true consumer surplus changes. Flexible model: $|h| = 20$, D_t included, $T = 1,000$. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

Merger Simulation: Merger evaluation represents a prominent counterfactual in applied work, and thus an important application of our method. Following the stan-

²¹For revenue predictions, MSE is computed as $MSE = \frac{1}{K} \sum_{k=1}^K (\hat{R}_k - R_k)^2$, where $\hat{R}_k$ and R_k are predicted and true revenues at tax rate k , and K is the number of tax rates evaluated. We use MSE rather than RMPSE for Laffer curves because revenue approaches zero at high rates.

dard approach outlined in Example 2 of Section 2, we model mergers as changes in the ownership matrix $\mathcal{H}_t$ while maintaining all products in the market. Our merger simulation environment features two market structures: half contain three multi-product firms (with 6, 5, and 4 products, respectively), while the other half contains four firms (one each with 5 and 4 products, and two with 3 products each). We introduce additional variation by randomly dropping 0–5 products per market, yielding 10–15 active products. The merger counterfactual consolidates the two smaller firms in four-firm markets. This setup makes our counterfactual relatively close to the support of the data, allowing the flexible model to learn how ownership affects pricing.

Figure 3 presents the mean prediction errors for consumer surplus changes from mergers. Under the Bertrand DGP, the flexible model achieves an RMPSE between 1.44 and 1.94, meaning the average prediction deviates from the truth by roughly 1.5–2% of the true counterfactual value, substantially outperforming the misspecified monopoly and perfect competition models (both 7.90–10.12). The performance advantage is similar under the profit-weight DGP, where the flexible model’s RMPSE between 1.53 and 1.94 is roughly half that of the misspecified parametric models (3.26–3.97 for Bertrand, 4.55–5.21 for monopoly and perfect competition). Figure 3 also reports the TF-RF row, with CS RMPSEs between 7.12 and 12.11 under Bertrand and between 17.65 and 25.80 under profit-weight.

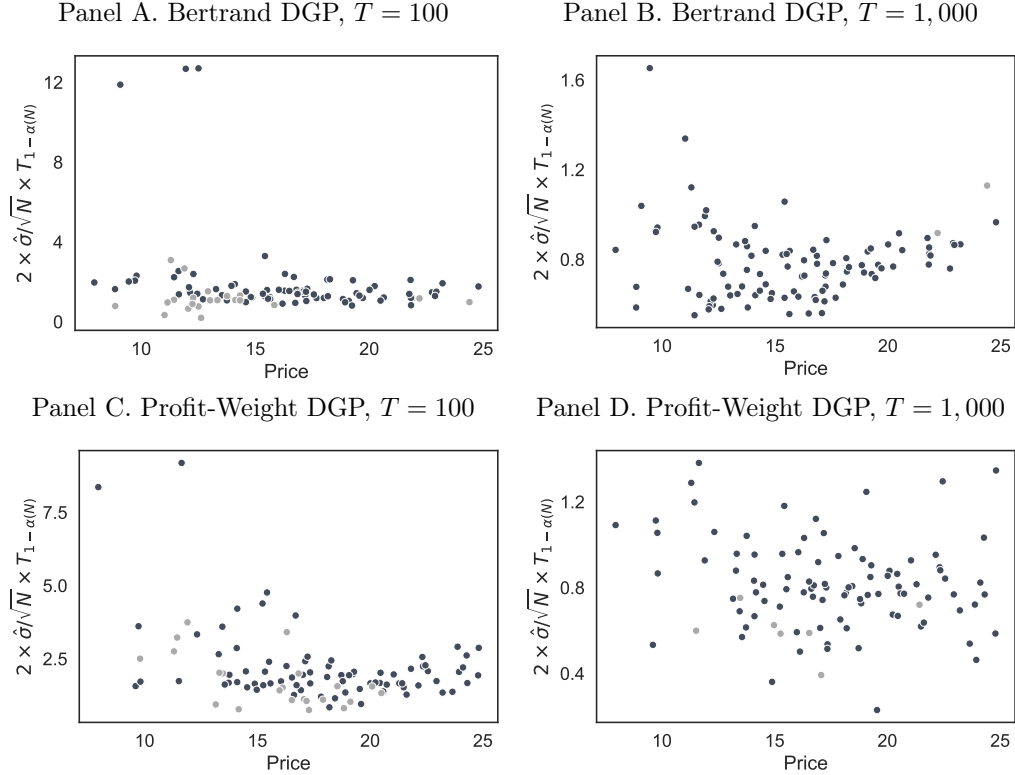
Other Counterfactuals: Beyond the three primary counterfactuals above, the Online Supplemental Materials (Section S.3.2) present results for product exit counterfactuals and cost shifter changes. The flexible model maintains prediction accuracy even for cost changes of 50–100% beyond the training support (Supplemental Figure S.3.2), confirming that $\hat{h}$ provides a general-purpose tool for counterfactual analysis.

5.6 Quantification of Uncertainty in Prediction

Beyond point predictions, quantifying uncertainty is essential for policy decisions. We evaluate the finite-sample performance of our inference procedure using a product exit counterfactual, where one product is removed and we predict the resulting equilibrium prices with 95% confidence intervals. Figure 4 displays confidence interval widths (computed using the delta method with Bonferroni correction) against predicted prices across 100 test markets, examining both small ($T = 100$) and typical

($T = 1,000$) sample sizes under Bertrand and profit-weight DGPs.

FIGURE 4: Inference on Counterfactual Merger Simulation Prices



Panel E. Coverage Probability (%)

	$T = 100$		$T = 1,000$	
	A. Bertrand	B. Profit-Weight	C. Bertrand	D. Profit-Weight
Coverage	79%	78%	98%	93%

Notes: Each point represents the Bonferroni-corrected 95% confidence interval width for a single product's counterfactual price following a product exit, across 100 test markets. Panel E reports coverage rates. Flexible model: $|h| = 20$, D_t included. See Supplemental Figure S.3.11 for results without D_t .

The results demonstrate strong inference properties that improve markedly with sample size. With $T = 1,000$ markets, our procedure achieves 93–98% coverage rates, closely matching the nominal 95% level, while confidence interval widths shrink substantially compared to $T = 100$. The heteroscedastic pattern (wider intervals at price extremes) reflects data density in the training sample and captures greater uncertainty where fewer observations are available. These results confirm that our inference procedure provides reliable uncertainty quantification in finite samples.

6 Empirical Application

As an application of our method, we predict the effects of a merger in the U.S. airline industry. The airline industry has received substantial attention from the IO literature (starting with [Berry, 1992](#); [Berry, Carnall, and Spiller, 2006](#)) given the rich available data and significant consolidation over the last several decades. The retrospective studies of large mergers have had mixed results ([Peters, 2006](#)), potentially linked to non-Bertrand conduct (see, e.g., [Ciliberto and Williams, 2014](#), for evidence of non-competitive conduct). Our method applies well here, given the large amount of data and variation in market structure.²²

6.1 Background and Data

We construct a database of the U.S. airline industry during the period 2005–2019. We use the 10 percent sample of purchased airline tickets from the well-known Airline Origin and Destination Survey (DB1B) database released by the U.S. Department of Transportation. A market is defined as a pair of cities, regardless of the flight direction. We match cities to Metropolitan Statistical Areas (MSA) and collect data on the populations of these MSAs from the Bureau of Economic Analysis. The geometric mean of endpoint populations is used as a measure of the market size. A product is a one-way trip that services a particular city-pair and is defined at the carrier-market-quarter level. Additional details on the construction of the data can be found in [Appendix E.1](#).

The U.S. airline industry has experienced substantial consolidation in the last two decades with legacy carriers and low-cost airlines participating in large mergers. The earliest merger in our data is the Delta-Northwest merger in 2008. The merger was announced on April 14, 2008, and was approved on October 29, 2008, after roughly six months of review by the U.S. Department of Justice (DOJ). Given the limited overlap between the merging airlines’ operations, the merger was perceived as having a modest impact on competition. The second merger included is the United-Continental

²²We emphasize the illustrative nature of our application. Recent papers have highlighted the dynamic nature of pricing and demand in this market (e.g., [Williams, 2022](#); [Hortaçsu, Natan, Parsley, Schweg, and Williams, 2024](#)), and the role of endogenous network structure (e.g., [Bontemps, Gualdani, and Remmy, 2023](#); [Yuan and Barwick, 2024](#)). We abstract away from these elements to keep our application close to standard merger simulations.

merger in 2010. The DOJ approved the merger after four months of review on August 27, 2010. As a condition of approval, the merged entity was required to lease slots to Southwest at Newark Liberty Airport in New Jersey. Finally, we consider the controversial merger of American Airlines and US Airways. The last of the “mega-mergers” that involved two airlines, the deal was announced on November 12, 2013. According to the settlement terms, the merged entity was required to divest slots at several major airports, most prominently at Ronald Reagan Washington National Airport and New York’s LaGuardia Airport. More recently, outside our analysis, Alaska Airlines acquired Virgin America and Hawaiian Airlines, and a federal court blocked JetBlue’s attempted acquisition of Spirit.

6.2 Demand and Supply Estimation

Demand Estimation: We follow [Berry and Jia \(2010\)](#) in adopting a nested logit demand model. We briefly summarize the model here and provide additional details in [Appendix E.2](#). Product characteristics include average fares, the share of nonstop flights, the average distance in thousands of miles, and a squared distance term. We restrict our attention to the major carriers, controlling for the number of fringe firms to capture variation in market structure over time across origin-destination pairs. We include origin-destination fixed effects. Our nesting structure includes all inside goods in one nest. We include instruments to handle endogeneity issues for prices and nests. We use BLP-style instruments: average rival distance, average number of markets served by rivals, and the number of rival carriers.

The results for demand estimation are reported in [Table E2](#). In line with the previous literature, we find that consumers prefer a higher share of nonstop flights and incur disutility from more miles traveled. There is strong within-nest substitution. The median own-price elasticity of -5.17 matches the literature well. Given the simplicity of the nested logit specification and the large sample size available in airline data, first-step demand estimation error is unlikely to materially affect our supply estimates.

Pre-merger Supply Estimation: We consider two models of conduct for our supply specifications: (i) Bertrand pricing with constant marginal costs and (ii) a flexible supply function as described in [Section 3](#). Both specifications satisfy [Assumption 6](#). In

this section, we focus on the flexible specification and relegate details of the Bertrand specification to Appendix E.3.

We estimate the flexible supply model with the VMM estimator described in Section 4. We include in the supply function market shares and the average distance in thousands of miles as an observable cost shifter. We also include origin-destination fixed effects. We instrument the endogenous market shares with BLP-style instruments: average rival distance, average number of markets served by rivals, and the number of rival carriers. Additionally, we include own-product characteristics that do not directly impact marginal costs – the share of nonstop flights and squared average distance in thousands of miles – as excluded instruments. We stratify the pre-merger data by market and split it into 80% of markets for training, leaving the remaining 20% of markets as a hold-out test sample to evaluate the fit.

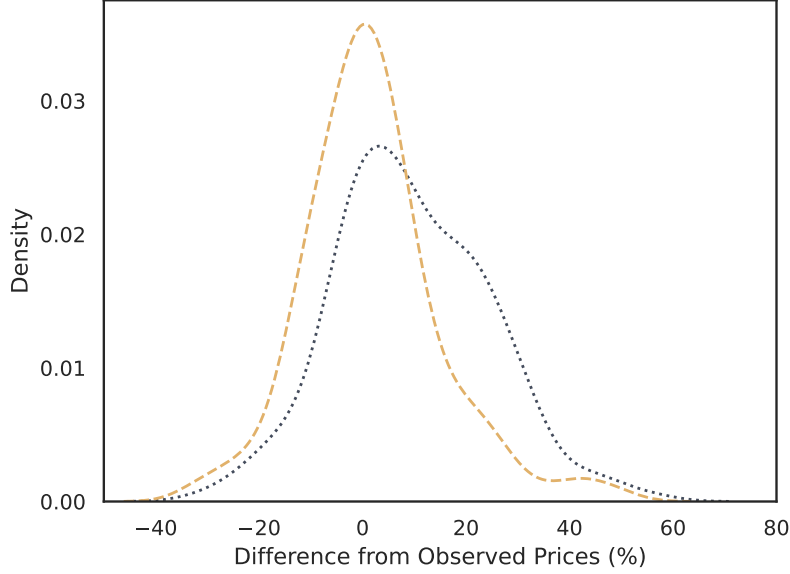
Having estimated the two models, we can evaluate their ability to explain the variation in prices in the pre-merger data. To do so, we compare the variance of the model-implied estimates of the unobservable cost shock. The flexible supply function estimated with VMM significantly outperforms the parametric supply model as it implies a variance in ω that is 44% smaller than the variance implied by Bertrand conduct with constant marginal cost. This margin is similar in the training data and a test sample.

6.3 Merger Simulation Results

We examine the merger of American Airlines and US Airways in our counterfactuals. We focus on markets with three firms in the pre-merger period and two firms in the post-merger period. We compare the predictions of the models to the true post-merger prices. The observed price differences are presented in Figure E1. In constructing the counterfactual, we hold the demand and cost unobservables fixed at their pre-merger estimated values, changing only the ownership structure to reflect common ownership of American Airlines and US Airways products.

Figure 5 plots the distribution of percentage differences between predicted and observed prices for the two methods, the flexible supply model and the Bertrand with constant marginal cost model, in the post-merger period. The flexible model estimated with VMM is centered at zero, with a large fraction of predicted prices falling within 20% of realized prices, and a passenger-weighted MSE of 144.06. Instead, the

FIGURE 5: Merger Simulation Results



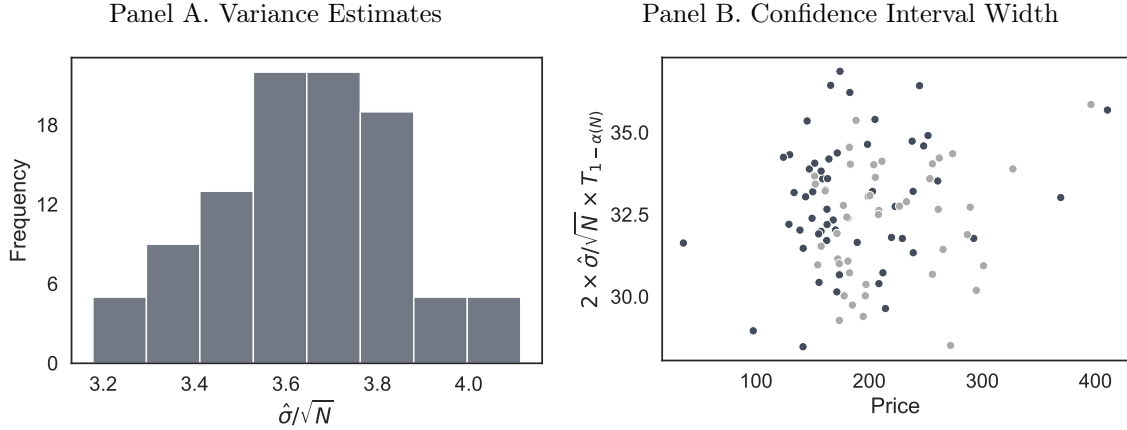
Notes: The figure reports merger simulation results for the flexible model estimated with VMM (in yellow) and the standard merger simulation model (in blue).

standard merger simulation method systematically over-predicts changes (similar to the findings in [Bhattacharya et al., 2025](#)) and has an overall passenger-weighted MSE of 817.75. In sum, our flexible model substantially outperforms the standard simulation toolkit when predicting post-merger prices of the American Airlines-US Airways merger.

Quantifying Uncertainty: Finally, we quantify the uncertainty of our predictions in the merger simulation exercise. We follow Algorithm 1 to construct the confidence intervals. Notably, we use a more conservative Bonferroni correction for ease of exposition, allowing us to present the results with a single set of bounds for each point. We construct bounds for all points in the sample selected for merger simulation.

The results are presented in Figure 6. We present the variance estimates in Panel A and the total width of the confidence interval as a summary of estimation uncertainty in Panel B. The predictions of the inference exercise show that uncertainty is roughly constant with price levels. The widths are moderate relative to the level of prices, suggesting that estimation uncertainty is not the main source of dispersion in post-merger prediction errors, even with a high-dimensional markup function.

FIGURE 6: Inference Results



Notes: Inference on predicted prices following the American Airlines-US Airways merger. Panel A: estimated variance of predicted prices. Panel B: Bonferroni-corrected confidence interval width versus observed post-merger prices. Dark points indicate observed prices within the confidence interval (57%); light points fall outside (43%).

7 Conclusion

This paper demonstrates how ML/AI methods can enhance counterfactual prediction while maintaining (nonparametric) economic structure. We develop a flexible supply function that nests standard oligopoly models without imposing specific assumptions about conduct or costs, and estimate it using neural VMM (Bennett and Kallus, 2023), which provides a practical solution to the curse of dimensionality that has limited nonparametric approaches in IO. Our Monte Carlo evidence establishes that the approach delivers predictive accuracy close to correctly specified models, scales to realistic market sizes, extrapolates to policy counterfactuals outside the training support, and provides reliable uncertainty quantification. Our application to airline mergers demonstrates that these advantages matter in practice: the flexible model delivers a fivefold improvement in prediction accuracy over standard Bertrand assumptions, suggesting that misspecified conduct assumptions may systematically bias policy recommendations. The method is portable across many applications and easy to implement. Appendix F presents a stylized example illustrating how researchers can use the method.

Our results point to several directions for future research. Because the flexible supply function is estimated without imposing a model of competition, our framework

could serve as a discovery tool: comparing the estimated $\hat{h}$ to the predictions of standard conduct models may reveal competitive behaviors not captured by any model on the conventional menu. A natural extension concerns the role of demand. Our current approach conditions on a first-step demand estimate; misspecification of the demand system could propagate to supply estimates and counterfactual predictions. Developing joint nonparametric estimation of demand and supply, where both sides of the market are learned flexibly from data, would mitigate this concern and is a promising direction for future work.

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A Identification of the Supply Function: Proofs and Additional Discussion

A.1 Proof of Theorem 1

The proof adapts the standard NPIV logic from [Newey and Powell \(2003\)](#) to our manifold setting. Under Assumptions 1-8, suppose two functions h_j^1 and h_j^2 both satisfy the moment condition in Equation (3). Then:

$$\mathbb{E}[(h_j^1 - h_j^2)(s_t, D_t, w_{jt}) \mid z_{jt}, w_{jt}] = 0.$$

By the manifold completeness condition (Assumption 8), this implies $(h_j^1 - h_j^2) = 0$ almost surely on $\mathcal{M}$. Therefore, h_j is uniquely identified on the manifold. This follows directly from Proposition 2.1 in [Newey and Powell \(2003\)](#), applied to the manifold $\mathcal{M}$ rather than the full space.

A.2 Understanding Manifold Completeness

The manifold completeness condition requires that instrumental variation be rich enough to identify the supply function on all feasible combinations of (s_t, D_t) . To understand why this condition is reasonable, consider the necessary rank condition that makes completeness meaningful.

Given the known demand system and index structure, the manifold $\mathcal{M}$ is implicitly defined by the constraint $D_t = D(s_t, \delta_t, x_t^{(2)})$ where $\delta_t = x_t^{(1)} + \xi_t$. For identification on this manifold, we need the rank condition:

$$\text{rank} \left[\frac{\partial \text{vec}(D(s_t, \delta_t, x_t^{(2)}))}{\partial (\delta_t, \text{vec}(x_t^{(2)}))} \right] = JK.$$

This rank condition ensures that variations in the demand index δ_t and characteristics $x_t^{(2)}$ generate sufficient independent directions of movement in the demand derivatives. Without this property, different supply functions could be observationally equivalent, making completeness vacuous. The rank condition is not a separate assumption but rather clarifies what we mean by completeness in this setting. It guarantees that variation in $x_t^{(1)}$ shifts the demand index $\delta_t = x_t^{(1)} + \xi_t$, variation in $x_t^{(2)}$ directly affects demand, and together with rival cost shifters $w_{-j,t}$ affecting equilibrium shares, these instruments span the manifold.

The key insight is dimensional reduction through the manifold structure. While (s_t, D_t) live in $\mathbb{R}^{J+J^2}$, the true endogenous variation comes only from (s_t, δ_t) , which has dimension $2J$. The exogenous characteristics $x_t^{(2)}$ can be conditioned upon, leaving only $2J$ dimensions of endogenous variation that need to be instrumented. This represents a dramatic reduction from the $J + J^2$ dimensions of the full space. For example, with $J = 30$ products, standard completeness would require instruments for variation in $30 + 900 = 930$ dimensions, while our approach only needs to handle 60 dimensions of endogenous variation coming from (s_t, δ_t) . This dimensional reduction, from $J + J^2$ to $2J$, makes nonparametric identification feasible even in markets with many products.

A.3 Comparison with Berry and Haile (2014)

Our identification strategy is related to results in Section 4.4 in [Berry and Haile \(2014\)](#), who identify the inverse supply function $\pi^{-1}(s_t, p_t)$ mapping market shares and prices to values of a cost index $\kappa_{jt} = w_{jt}^{(1)} + \omega_{jt}$.

The approaches share several key features. Both use exclusion restrictions, where some product characteristics are excluded from costs, and both require completeness conditions, although on different spaces. However, we identify $h_j(s_t, D_t, w_{jt}) = \Delta_j + c_j$ directly, while [Berry and Haile \(2014\)](#) identify the inverse π^{-1} . While both formulations of supply may be used for some counterfactuals, our direct estimation of the supply function lends itself to further decomposition into markups and cost, an approach that we discuss in the Online Supplemental Materials (Section [S.1.1](#)).

B On Deep Neural Networks Architectures

In a nonparametric regression framework, [Schmidt-Hieber \(2020\)](#) demonstrates that estimators based on sparsely connected deep neural networks (DNNs) with ReLU activation functions attain the minimax rate of convergence. In Section 5, the author further shows that nonparametric regression using wavelet bases achieves only suboptimal rates. Drawing on these insights from [Schmidt-Hieber \(2020\)](#), we provide an explanation for why our DNN-based VMM estimator outperforms its series-based NPIV counterpart.

Let $\sigma(x) = \max\{0, x\}$. For $\mathbf{v} = (v_1, \dots, v_r) \in \mathbb{R}^r$, $r \in \mathbb{N}$, define:

$$\sigma_{\mathbf{v}}(y_1, \dots, y_r) = (\sigma(y_1 - v_1), \dots, \sigma(y_r - v_r)).$$

A neural network with network architecture $(L, \mathbf{p})$, where $L > 0$ is the number of hidden

layers and $\mathbf{p} = (p_0, p_1, \dots, p_{L+1}) \in \mathbb{N}^{L+2}$ is a width vector, is a function of the form:

$$f : \mathbb{R}^{p_0} \rightarrow \mathbb{R}^{p_{L+1}}, \quad [0, 1]^d \ni \mathbf{x} \mapsto f(\mathbf{x}) = W_L \sigma_{\mathbf{v}_L} W_{L-1} \sigma_{\mathbf{v}_{L-1}} \dots W_1 \sigma_{\mathbf{v}_1} W_0 \mathbf{x}, \quad (\text{B1})$$

where W_i is a $p_{i+1} \times p_i$ weight matrix and $\mathbf{v}_i \in \mathbb{R}^{p_i}$. Define the set of s -sparse networks as:

$$\mathcal{F}(L, \mathbf{p}, s, F) = \left\{ f \text{ of the form (B1)} : \max_{j=0, \dots, L} \|W_j\|_\infty \vee |\mathbf{v}_j|_\infty \leq 1, \sum_{j=0}^L \|W_j\|_0 + |\mathbf{v}_j|_0 \leq s, \|f\|_\infty \leq F \right\}.$$

Also define the set of functions that can be expressed as compositions of some Hölder functions by:

$$\mathcal{G}(q, \mathbf{d}, \mathbf{t}, \beta, K) = \left\{ f = g_q \circ \dots \circ g_0 : g_\ell = (g_{\ell m})_m : [a_i, b_i]^{d_\ell} \rightarrow [a_{\ell+1}, b_{\ell+1}]^{d_{\ell+1}}, \right. \\ \left. g_{\ell m} \in \mathcal{C}_{t_\ell}^{\beta_\ell}([a_\ell, b_\ell]^{t_\ell}, K) \text{ for some } |a_\ell|, |b_\ell| \leq K \right\},$$

where $\mathcal{C}_r^\beta(D, K)$ denotes the set of real-valued functions defined on a domain $D \subset \mathbb{R}^r$ that belong to the Hölder space of smoothness β and have Hölder norm bounded by K . In the Proof of Theorem 1 in [Schmidt-Hieber \(2020\)](#), it was shown in Equation (26) and the following paragraph that:

$$\inf_{f^* \in \mathcal{F}(L, \mathbf{p}, s, F)} \|f^* - f_0\|_\infty^2 \leq C \max_{\ell=0, \dots, q} c^{-\frac{2\beta_\ell^*}{t_\ell}} n^{-\frac{2\beta_\ell^*}{2\beta_\ell^* + t_\ell}} \quad (\text{B2})$$

for the function $f_0 \in \mathcal{G}(q, \mathbf{d}, \mathbf{t}, \beta, K)$ and a constant $C > 0$, where t_ℓ is the maximal number of variables on which each of the $g_{\ell m}$ depends. It is important to note that this is a purely approximation-theoretic result; no assumptions are made regarding the structure of the error term in the regression model. Moreover, the ambient dimension d of the function f_0 does not appear directly in the convergence bound. Instead, the rate depends on the intrinsic dimensions t_ℓ of the latent functions within the compositional structure of f_0 .

We observe that each component of the markup function $f_0(\cdot)$ arising from Bertrand competition can be well approximated by a deep neural network (DNN), as it belongs to the function class $\mathcal{G}(q, \mathbf{d}, \mathbf{t}, \beta, K)$. To see this, note that following Section 2.2 of [Magnolfi, Quint, Sullivan, and Waldfogel \(2022\)](#), the j -th component of the (recentered) Bertrand markup function can be written as:

$$f_{0j}(\mathbf{p}, \mathbf{c}, \Omega, S_p) = p_j - c_j - \left[(\Omega \odot S_p')^{-1} s \right]_j,$$

where $\mathbf{p}$ and $\mathbf{c}$ are vectors of prices and marginal costs, respectively, Ω is the ownership matrix, and S'_p denotes the matrix of price derivatives of market shares. Typically, Ω exhibits a sparse block structure, which implies that $(\Omega \odot S'_p)^{-1}$ also inherits sparsity. This structural sparsity corresponds to a form of sparse tensor decomposition, as considered in Equation (14) of [Schmidt-Hieber \(2020\)](#), and hence places f_0 within the compositional Hölder class $\mathcal{G}(q, \mathbf{d}, \mathbf{t}, \boldsymbol{\beta}, K)$. The implication is that DNNs can approximate such functions at the faster rate given in Equation (B2), which depends on the latent dimensions t_ℓ , rather than the ambient dimensionality d typically governing nonparametric approximation rates.

Although series-based estimators also rely on the idea that an unknown function can be approximated by a linear combination of a finite number of basis functions, they require an explicit choice of basis. This choice can significantly influence the quality of approximation and, consequently, the convergence rate, unlike in the case of DNNs, which are data-adaptive and less reliant on prior structural assumptions.

Wavelet series, for instance, are well studied and known to possess attractive theoretical properties. However, ([Schmidt-Hieber, 2020](#), Section 5) shows that even for relatively simple target functions, such as additive models of the form:

$$f_0(\mathbf{x}) = h(x_1 + \cdots + x_d), \quad \text{with } h \in C_1^\alpha([0, d], K),$$

series approximations using wavelet bases may still attain suboptimal rates. Specifically, in the proof of Theorem 4, it is shown that for any $0 < \alpha \leq 1$ and Hölder radius $K > 0$, the following lower bound on the approximation error holds:

$$\sup_{h \in C_1^\alpha([0, d], K)} \|f^* - f_0\|_\infty^2 \geq C n^{-2\alpha/(2\alpha+d)},$$

for some constant $C > 0$, where f^* is any function constructed from compactly supported wavelet bases. We anticipate that a similar result can be established for the more nonlinear structures present in the current formulation of the markup function.

Recent advances in approximation theory highlight the advantages of DNNs over traditional series-based estimators in nonparametric settings. Unlike series-based methods, which rely on a pre-specified basis and may suffer from suboptimal convergence, DNNs adapt to the structure of the target function and can achieve faster approximation rates. This advantage is especially relevant for structural models such as those arising from supply side competition, where the markup function exhibits compositional and sparse structures well suited to DNN approximation. These insights suggest that DNN-based VMM estimators extend these benefits to more complex, nonlinear formulations of the markup function.

B.1 Reduced-Form Transformer Benchmark

We document the TF-RF benchmark used in Sections 3.3 and 5.

Architecture. The benchmark uses a set-transformer (Lee et al., 2019; Vaswani et al., 2017), an architecture that operates directly on variable-sized sets. Each product in a market is encoded as a feature vector (a token), and the network maps the set of product tokens to the vector of product prices. The architecture is permutation equivariant: relabeling the products in a market relabels the outputs and changes nothing else. This embeds the symmetry restriction of Assumption 6 directly in the architecture; it could be relaxed by allowing product-specific features.

Tokenization. Each product j in market t contributes a token containing (x_{jt}, w_{jt}) and firm identity. The token carries no equilibrium shares, no demand derivatives D_t , and no baseline price; the predictor sees only exogenous variables. A second token (the intervention token) carries information about the counterfactual: the type of intervention (a merger, a tax change, or a regulation) and its parameters (the new ownership structure, the tax rate, the shifted characteristic value).

Network specification. The network has approximately 20,000 parameters and trains on a CPU in minutes. We use two stacked attention blocks of four heads each, with hidden dimensions $d_{\text{model}} = 64$ and $d_{\text{ff}} = 128$. Training uses the Adam optimizer with a 5% validation set for early stopping.

Estimation protocol. Each TF-RF network is trained with the hyperparameters above and evaluated on a hold-out test set. The reported $[Q_{10}, Q_{90}]$ brackets aggregate 50 runs per cell, following the convention established in Section 5.

What the architecture lacks. The TF-RF predictor lacks three ingredients that the structural approach uses: it does not impose the moment condition $\mathbb{E}[\omega \mid z] = 0$ (so there are no instruments); it does not solve an equilibrium fixed point (the predicted price comes from a single network evaluation rather than from $p = c + \eta(p; s(p))$); and it does not invert demand (shares do not enter the predictor). Thus, TF-RF is the cleanest empirical test of the non-identification result of Berry and Benkard (2006): the predictor sees only exogenous variables whose mapping to equilibrium prices is the object that is not nonparametrically identified.

C Omitted Details for Quantification of Uncertainty

The following theorem provides the main theoretical foundation for our inference procedure. Following [Bennett and Kallus \(2023\)](#), we assume throughout this section that the true parameter value θ_0 lies in Θ , a compact subset of a finite-dimensional Euclidean space.²³

Theorem 2. *Let $\tilde{\theta}_N \xrightarrow{p} \theta_0$. Suppose that, for each given $\mathcal{H}$, the map $h : \Theta \rightarrow \mathbb{R}^{\bar{d}}$ is differentiable at $\theta_0 \in \Theta$. Under the regularity conditions imposed by Theorems 2–3 in [Bennett and Kallus \(2023\)](#), we have:*

$$\left\{ \nabla_{\theta'} h(\theta_0) \Omega_0^{-1} \nabla_{\theta'} h(\theta_0)' / N \right\}^{-1/2} (h(\hat{\theta}_N) - h(\theta_0)) \xrightarrow{d} N(0, I). \quad (\text{C1})$$

Proof. Suppose $\tilde{\theta}_N \xrightarrow{p} \theta_0$ and the regularity conditions from Theorems 2–3 in [Bennett and Kallus \(2023\)](#) hold. This implies:

$$\{\Omega_0/N\}^{-1/2} (\hat{\theta}_N - \theta_0) \xrightarrow{d} N(0, I),$$

where $\Omega_0 = \mathbb{E} \left[\mathbb{E}[\nabla_{\theta} \omega(\theta_0) \mid z, w] \mathbb{E}[\omega(\theta) \omega(\theta)' \mid z, w]^{-1} \mathbb{E}[\nabla_{\theta} \omega(\theta_0) \mid z, w] \right]$. Now apply the function h to the set of $\bar{d}$ observations. The result follows directly from the delta method. $\square$

Next, we extend the asymptotic result to cover smooth functions of prices.

Assumption 9. (*Smooth Counterfactuals*) The counterfactual map F is continuous and once differentiable in prices such that $\nabla_p F$ (or equivalently $\nabla_h F(h(\theta))$) exists.

Assumption 9 is satisfied by a large range of counterfactuals, e.g., logit shares, consumer surplus, and government revenue. We can construct confidence intervals by applying the delta method to the specific composite function $F(h(\theta))$.

Theorem 3. *Under Assumption 9 and the regularity conditions in Theorems 2–3 in [Bennett and Kallus \(2023\)](#), we have:*

$$\left\{ \nabla_h F(h(\theta_0)) \nabla_{\theta} h(\theta_0) \Omega_0^{-1} \nabla_{\theta} h(\theta_0)' \nabla_h F(h(\theta_0))' / N \right\}^{-1/2} (F(h(\hat{\theta}_N)) - F(h(\theta_0))) \xrightarrow{d} N(0, I). \quad (\text{C2})$$

²³Compactness is required only for establishing asymptotic normality, not for consistency.

Proof. The numerical delta method in [Hong et al. \(2015\)](#) implies Equation (C1). Applying the delta method under Assumption 9 to the composite function $F(h(\theta))$ to Equation (C1), we find Equation (C2). $\square$

Taking $\beta = \nabla_h F(h_x(\theta_0)) \nabla_\theta h_x(\theta_0)$ for some selected set of products x , Lemma 9 in [Bennett and Kallus \(2023\)](#) implies that the asymptotic variance for $F(h_x(\theta))$ in Equation (C2) can be estimated using the same method as described in Equation (5). We can construct simultaneous confidence intervals for many predictions, e.g., market shares, using Algorithm 1 or for singletons, e.g., total consumer surplus, directly with Equation (5). Below, we present a series of corollaries for specific counterfactuals.

Corollary 1. *Suppose we would like to conduct inference on counterfactual market share for product j with the map $F(h(\theta)) = s_j(h(\theta))$. We have:*

$$\left\{ D_j(h(\theta_0)) \nabla_\theta h(\theta_0) \Omega_0^{-1} \nabla_\theta h(\theta_0)' D_j(h(\theta_0))' / N \right\}^{-1/2} (F(h(\hat{\theta}_N)) - F(h(\theta_0))) \xrightarrow{d} N(0, I),$$

where D_j is the row corresponding to product j in the demand derivative matrix evaluated at counterfactual prices $h(\theta)$.

Proof. For a generic demand system, we find that $\nabla_h F(h(\theta)) = D_j(h(\theta))$. For example, take logit demand. Using the functional form of logit, we find $[\nabla_h F(h(\theta))]_k$ at index k as:

$$[\nabla_h F(h(\theta))]_k = [D_j(h(\theta))]_k = \begin{cases} -\alpha_p s_j(h(\theta))(1 - s_j(h(\theta))) & j = k \\ \alpha_p s_j(h(\theta)) s_k(h(\theta)) & j \neq k \end{cases}$$

Stacking the elements into a $1 \times J$ matrix, this is equivalent to row j of the derivative matrix of demand evaluated at the counterfactual $h(\theta)$, defined above as $D_j(\cdot)$. Thus, more generically, $\nabla_h F(h(\theta)) = D_j(h(\theta))$. The result follows immediately from Theorem 3. $\square$

Corollary 2. *Denote $\mathcal{G} = \mathcal{G}_1 \cup \dots \cup \mathcal{G}_M$ as the union of product sets across markets $m = 1, \dots, M$. Suppose we would like to conduct inference on the counterfactual total quantity across markets with the map $F(h(\theta)) = \sum_{j \in \mathcal{G}} s_j(h(\theta))$. We have:*

$$\left\{ 1'_{|\mathcal{G}|} D(h(\theta_0)) \nabla_\theta h(\theta_0) \Omega_0^{-1} \nabla_\theta h(\theta_0)' D(h(\theta_0))' 1_{|\mathcal{G}|} / N \right\}^{-1/2} (F(h(\hat{\theta}_N)) - F(h(\theta_0))) \xrightarrow{d} N(0, I),$$

where $\nabla_h F(h(\theta)) = 1'_{|\mathcal{G}|} D(h(\theta))$. $D(\cdot)$ is stacked block-diagonal across markets and evaluated at counterfactual prices $h(\theta)$.

Proof. Taking the gradient and borrowing notation from the previous proof, we find:

$$[\nabla_h F(h(\theta))]_j = \sum_{k \in \mathcal{G}} [D(h(\theta))]_{kj}.$$

Notice that, by market independence, $[D(h(\theta))]_{kj} = 0$ whenever j and k are in different markets. Stacking the gradients $D(\cdot)$ block-diagonal across markets, we have a $1 \times |\mathcal{G}|$ matrix $\nabla_h F(h(\theta)) = 1'_{|\mathcal{G}|} D(h(\theta))$. The result follows immediately from Theorem 3. $\square$

Corollary 3. *Suppose we would like to conduct inference on total consumer surplus across markets $m = 1, \dots, M$ from logit demand with the map*

$$F(h(\theta)) = -\frac{1}{\alpha_p} \sum_m \log \left(1 + \sum_{j \in \mathcal{G}_m} \exp(-\alpha_p h_j(\theta) + x_j \beta + \xi_j) \right).$$

We have:

$$\left\{ s(h(\theta_0))' \nabla_\theta h(\theta_0) \Omega_0^{-1} \nabla_\theta h(\theta_0)' s(h(\theta_0)) / N \right\}^{-1/2} (F(h(\hat{\theta}_N)) - F(h(\theta_0))) \xrightarrow{d} N(0, I),$$

where $s(h(\theta))$ are market shares evaluated at counterfactual prices $h(\theta)$.

Proof. Taking the gradient of the map F with respect to h , we find the j -th element as:

$$[\nabla_h F(h(\theta))]_j = \frac{\exp(-\alpha_p h_j(\theta) + x_j \beta + \xi_j)}{1 + \sum_{k \in \mathcal{G}_{m(j)}} \exp(-\alpha_p h_k(\theta) + x_k \beta + \xi_k)} = s_j(h(\theta)).$$

Notice that, by market independence, gradients with respect to prices in other markets are zero. Stacking the gradients across markets, we have a $1 \times |\mathcal{G}|$ matrix of transposed market shares $s(\cdot)'$ evaluated at the counterfactual $h(\theta)$. The result follows immediately from Theorem 3. $\square$

Corollary 4. *Denote $\mathcal{G} = \mathcal{G}_1 \cup \dots \cup \mathcal{G}_M$ as the union of product sets across markets $m = 1, \dots, M$. Suppose we would like to conduct inference on government revenue from ad valorem taxes with tax shares a across markets with the map $F(h(\theta)) = \sum_{j \in \mathcal{G}} a_j h_j(\theta) s_j(h(\theta))$. We have:*

$$\left\{ G(h(\theta_0)) \nabla_\theta h(\theta_0) \Omega_0^{-1} \nabla_\theta h(\theta_0)' G(h(\theta_0))' / N \right\}^{-1/2} (F(h(\hat{\theta}_N)) - F(h(\theta_0))) \xrightarrow{d} N(0, I),$$

where $G(h(\theta)) = (a \odot s(h(\theta)))' + (a \odot h(\theta))' D(h(\theta))$, and $D(\cdot)$ is stacked block-diagonal by

markets.

Proof. Taking the gradient of the map F with respect to h , we find the j -th element as:

$$\begin{aligned} [\nabla_h F(h(\theta))]_j &= a_j[s_j(h(\theta)) + h_j(\theta)[\nabla_h s(h(\theta))]_{jj}] + \sum_{k \neq j} a_k h_k(\theta)[\nabla_h s_k(h(\theta))]_{kj} \\ &= a_j s_j(h(\theta)) + \sum_k a_k h_k(\theta)[D(h(\theta))]_{kj}. \end{aligned}$$

Notice that, by market independence, gradients with respect to h across markets are zero. Stacking the gradients $D(\cdot)$ block-diagonal across markets, we have:

$$G(h(\theta)) \equiv \nabla_h F(h(\theta)) = (a \odot s(h(\theta)))' + (a \odot h(\theta))' D(h(\theta)).$$

The result follows immediately from Theorem 3. □

Corollary 5. Denote $\mathcal{G} = \mathcal{G}_1 \cup \dots \cup \mathcal{G}_M$ as the union of product sets across markets $m = 1, \dots, M$. Suppose we would like to conduct inference on government revenue from unit taxes τ across markets with the map $F(h(\theta)) = \sum_{j \in \mathcal{G}} \tau_j s_j(h(\theta))$. We have:

$$\left\{ \tau' D(h(\theta_0)) \nabla_{\theta} h(\theta_0) \Omega_0^{-1} \nabla_{\theta} h(\theta_0)' D(h(\theta_0))' \tau / N \right\}^{-1/2} (F(h(\hat{\theta}_N)) - F(h(\theta_0))) \xrightarrow{d} N(0, I),$$

where $\nabla_h F(h(\theta)) = \tau' D(h(\theta))$ and $D(\cdot)$ is stacked block-diagonal across markets.

Proof. Taking the gradient of the map F with respect to h , we find the j -th element as:

$$\begin{aligned} [\nabla_h F(h(\theta))]_j &= \tau_j[\nabla_h s(h(\theta))]_{jj} + \sum_{k \neq j} \tau_k[\nabla_h s_k(h(\theta))]_{kj} \\ &= \sum_k \tau_k[D(h(\theta))]_{kj}. \end{aligned}$$

Notice that, by market independence, gradients with respect to h across markets are zero. Stacking the gradients $D(\cdot)$ block-diagonal across markets, we have $\nabla_h F(h(\theta)) = \tau' D(h(\theta))$. The result follows immediately from Theorem 3. □

D Simulation Details and Additional Results

We discuss additional details concerning our simulation environments and computational details used to implement our Monte Carlo experiments. We proceed in five steps. First,

we characterize the differences between the three simulation environments both in terms of market structure and demand. Second, we discuss the supply-side models used to generate data in all three environments. Third, we provide details on estimating VMM and the parametric models of supply. Fourth, we explain how we compute counterfactuals under the estimated VMM and parametric model. Fifth, we discuss the steps to quantify uncertainty arising in the predicted counterfactuals. Table D1 shows which environments and exercises are used for each step.

TABLE D1: Simulation Design Summary

Exercise	Env.	T	Conduct	Cost
<i>Predictive Accuracy in Hold-Out Samples</i>				
Holdout Performance (Sec. 5.2)	Base	100/1K/10K	B, PW	C, E
<i>Scalability</i>				
High-Dimensional (Sec. 5.3)	HiDim	100/1K	B, PW	C
<i>Economic Interpretability</i>				
Pass-Through (Sec. 5.4)	Base	1K	B, PW	C
<i>Policy Counterfactuals</i>				
Characteristic Regulation (Sec. 5.5)	Base	1K	B, PW	C, E
Tax Policy (Sec. 5.5)	Base	1K	B, PW	C
Merger Simulation (Sec. 5.5)	Merger	1K	B, PW	C
<i>Uncertainty Quantification</i>				
Inference (Sec. 5.6)	Base	100/1K	B, PW	C

Notes: Env. = Environment: Base = Baseline (1–3 products); HiDim = High-dimensional (30 products); Merger = Merger-specific (10–15 products, multi-product firms). Conduct: B = Bertrand; PW = Profit-weight. Cost: C = Constant marginal cost; E = Economies of scale. Multiple values separated by commas indicate separate results for each specification.

D.1 Details on Constructing Simulation Environments

Baseline Environment: We generate samples with $T \in \{100, 1,000, 10,000\}$ markets. In each market, we start with three products and then randomly drop at most one product from each market. As we only consider single-product firms, this results in approximately half the markets in a dataset being duopoly markets and half being triopoly markets. For demand, we adopt a logit specification. Consumer i derives utility from product j in market

t according to:

$$u_{ijt} = \alpha_p p_{jt} + \beta x_{jt} + \xi_{jt} + \epsilon_{ijt},$$

where x_{jt} represents a constant and two observed product characteristics $x_{jt}^{(1)}$ and $x_{jt}^{(2)}$. ξ_{jt} captures unobserved quality, and ϵ_{ijt} follows a Type I extreme value distribution. $x_{jt}^{(1)}$ and $x_{jt}^{(2)}$ are independently drawn from a $U[0, 1]$ distribution. ξ_{jt} is drawn from a $U[0, 1]$ distribution and has correlation $\rho = 0.9$ with unobserved supply shocks ω_{jt} , which is also drawn from $U[0, 1]$. For Laffer curve counterfactuals, we augment the environment with variation in taxes. Crucially for identification of the flexible supply function, markets feature variation in both ad valorem and unit taxes. Ad valorem tax rates are drawn from $v_t \sim U[0, 0.6]$, while unit taxes follow $\tau_t \sim U[4, 8]$.

High-Dimensional Environment: To address concerns about the curse of dimensionality inherent in nonparametric methods, we implement a second environment inspired by the empirical setting in [Miller and Weinberg \(2017\)](#). Following the market structure of the U.S. beer market in that paper, this environment features 30 differentiated products offered across markets, an order of magnitude larger than our baseline setup and representative of product variety in many IO settings. To generate data, we start with five firms in every market, each producing six products. We then randomly drop up to ten products, so that the final datasets contain 20–30 products in each market. For demand, we adopt a nested logit demand system, which is more flexible than in our baseline environment. Specifically, we adopt an inside-outside nesting structure so that the utility that individual i receives from inside product j in market t is given by:

$$u_{ijt} = \alpha_p p_{jt} + \beta x_{jt} + \xi_{jt} + \zeta_{it} + (1 - \sigma)\epsilon_{ijt},$$

where x_{jt} represents a constant and a single observed product characteristic $x_{jt}^{(1)}$. ξ_{jt} captures unobserved quality, ζ_{it} captures the random taste for inside products following the Cardell distribution [Cardell \(1997\)](#), and ϵ_{ijt} follows a Type I extreme value distribution. To best match the environment in [Miller and Weinberg \(2017\)](#), we treat $x_{jt}^{(1)}$ as analogous to their month-product fixed effect, setting $\beta = 1$ and drawing $x_{jt}^{(1)}$ from $N(0, 0.2)$, which approximates the empirical distribution of their estimated month-product fixed effects. We also follow [Miller and Weinberg \(2017\)](#) in setting $\alpha_p = -0.0887$ and $\sigma = 0.83$. The unobserved demand and cost shocks ξ_{jt} and ω_{jt} are jointly drawn from a $U[0, 1]$ distribution

with variance-covariance matrix:

$$V = \begin{bmatrix} 0.18 & 0.04 \\ 0.04 & 1.08 \end{bmatrix},$$

which matches the empirical variance-covariance matrix in [Miller and Weinberg \(2017\)](#).

Merger Simulation Environment: For merger simulation counterfactuals (see [Section 5.5](#)), we augment the high-dimensional environment to create richer variation in market structure and the ownership matrix $\mathcal{H}_t$. In doing so, the pre-merger data contain variation analogous to the merger we simulate, helping the flexible model learn how the proposed change in market structure affects equilibrium pricing. In 50% of the markets, there are three firms, one with six products, one with five products, and one with four products. In the other 50% of markets, there are four firms, one with five products, one with four products, and two with three products. We randomly drop up to five products from each market in the same manner as in the high-dimensional environment. Here, we use the exact demand system from the baseline environment.

D.2 Details on Supply-Side Models in the DGPs

Marginal Cost Specifications: In all three data-generating environments, we generically parameterize the marginal cost as

$$c_{jt} = \mathbf{w}_{jt}'\gamma + \lambda_0 s_{jt} + \lambda_1 s_{jt}^2 + \omega_{jt}.$$

In all simulations in the baseline and merger-simulation environments, $\mathbf{w}_{jt}$ contains a constant and two observed cost shifters, excluded from x_{jt} , which are drawn iid from a $U[0, 1]$ distribution. As discussed above, the unobserved cost shock ω_{jt} is drawn jointly with the unobserved demand shock ξ_{jt} from a standard bivariate uniform distribution with correlation coefficient $\rho = 0.9$ (the default in `pyblp`). In these environments, we set the parameters $\gamma = [3, 6, 4]$. The values of λ_0 and λ_1 allow us to control whether the supply model exhibits economies of scale. In DGPs with constant marginal cost, $\lambda_0, \lambda_1 = 0$; for economies of scale specifications, $\lambda_0 = 20$ and $\lambda_1 = -30$.

In simulations under the high-dimensional environment, we instead draw a single observed cost shifter from an exponential distribution with scale parameter 1.25, matching the empirical distribution of the observed cost shifter in [Miller and Weinberg \(2017\)](#). For $\mathbf{w}_{jt} = (1, \mathbf{w}_{jt}^{(1)})$, we set $\gamma = [6.809, 0.168]$. ω_{jt} is drawn jointly with ξ_{jt} according to the

uniform distribution described above so that the variance-covariance matrix approximates that in [Miller and Weinberg \(2017\)](#). In the high-dimensional environment, we only consider constant marginal costs so that $\lambda_0, \lambda_1 = 0$.

Supply-side Models: For the supply side, the generic first-order conditions generating prices can be expressed as:

$$p_{jt} = (\mathcal{H}_t \odot D_t)^{-1} s_t + c_{jt}.$$

When the supply-side model generating the data involves Bertrand conduct, the (j, k) -th element of the ownership matrix $\mathcal{H}_t$ equals zero when products j and k are produced by different firms in market t and one otherwise. For profit-weight models, the (j, k) -th element of $\mathcal{H}_t$ equals κ when products j and k are produced by different firms in market t . In the baseline environment, $\kappa = 0.5$; in the high-dimensional and merger-simulation environments, we set $\kappa = 0.75$.

Supply-side Models for Laffer Curve Exercises: As discussed above, we modify the baseline environment when generating data to perform the Laffer curve counterfactual by incorporating market-level ad valorem tax rates (v_t) and unit taxes (τ_t) into our simulations. We impose unit taxes directly on firms, while ad valorem taxes are levied on consumers. Defining p_{jt} as the tax-inclusive price paid by consumers, the fraction of the consumer’s payment received by the firm is $\nu_t = 1/(1 + v_t)$. The augmented after-tax first-order conditions generating firms’ prices are given by:

$$\nu_t p_{jt} - \tau_t = \nu_t \Delta_{jt} + c_{jt}, \tag{D1}$$

where $\nu_t p_{jt} - \tau_t$ is the per-unit net revenue received by the firm for product j in market t .

D.3 Estimation Details

Variational Method of Moments Deep neural networks require the specification of a number of hyperparameters. Our implementation of VMM is no exception: we use two deep neural networks to fit our flexible supply model. We describe the tuning choices used in the paper. For many of the hyperparameters, we leave the defaults from [Bennett and Kallus \(2023\)](#). We use the Optimistic Adam (OAdam) algorithm to train the deep neural networks, setting the learning rate to $\eta = 5 \times 10^{-4}$ and the momentum decay parameters to $\beta = [0.5, 0.9]$. We do not perform gradient descent by batching markets, instead opting to

include all markets in a single batch.²⁴ During training, we track loss for a smaller hold-out validation sample. Throughout, we implement early stopping by saving the weights of the deep neural networks corresponding to the lowest validation loss.²⁵ We do not include a regularization term R , leaving any regularization to our early stopping procedure. While we experiment with the dimensionality of the primary deep neural network h , we fix the dimensionality of the deep neural network f to that of [Bennett and Kallus \(2023\)](#): two hidden layers, the first with 50 nodes and the second with 20 nodes. Both of our deep neural networks f, h are fully connected with rectified linear units (ReLU) activation functions.

In the training process, we require specifications of endogenous, exogenous, and instrumental variables. The endogenous variables include the vector of J_t market shares and the matrix of $\frac{J_t(J_t-1)}{2} + J_t$ demand derivatives for each market t . The exogenous variables include K^w own cost shifters w_{jt} , where K^w is the number of observable cost shifters. For instruments, we include three sets: (i) $K^x \times J_t$ own and rival product characteristics, where K^x is the number of product characteristics that enter demand and are excluded from cost; (ii) $K^w \times (J_t - 1)$ rival cost shifters; and (iii) $1 + K^x + K^w$ constructed BLP instruments for other products and rival firms. In conjunction with the manifold completeness assumption for identification, we maintain sufficient dimensionality of instruments for identification.

Parametric Models For each parametric model, we estimate the model by imposing a model of conduct to invert marginal costs and then project marginal costs on observable cost shifters. We walk through each of the models in turn, beginning with the simplest specifications. Under perfect competition, we assume that firms price at marginal cost, meaning prices are equal to marginal costs. We regress these costs on observable cost shifters to recover the implied parameters and residuals. Next, turning to price-setting models (Bertrand, profit-weight, and monopoly), we assume a model of conduct with its corresponding ownership matrix and use the first-order conditions to invert marginal costs (described above). Again, we regress these costs on observable cost shifters to recover implied parameters and residuals. In some specifications of the data-generating process, we include economies of scale. Under economies of scale, we include market shares and squared market shares in the regression. To address the endogeneity present in the problem, we instrument these two variables with own product characteristics and squared own product characteristics under the assumption that they are excluded from the marginal cost. We recover the implied parameters and residuals, as before.

²⁴We found that this improved speed while leaving performance relatively unaffected.

²⁵We additionally define an ex ante early stopping rule but find that performance is comparable.

D.4 Hold-out Sample, Cost Pass-Through, and Counterfactual Performance

We now turn to our evaluation of the parametric and flexible models. Before doing so, it is useful to establish the computation of equilibrium prices. For parametric model m , we solve the fixed point in prices using Equation (D2) below:

$$\tilde{\nu}_t \tilde{p}_{jt} - \tilde{\tau}_t = \tilde{\nu}_t \tilde{\Delta}_j^m(\tilde{p}_t, \tilde{D}(\tilde{p}_t, \tilde{x}_t, \tilde{\xi}_t), \tilde{\mathcal{H}}_t) + \tilde{c}_j^m(\tilde{q}_t, \tilde{w}_{jt}, \tilde{\omega}_{jt}), \quad j = 1, \dots, J, \quad (\text{D2})$$

where variables with tildes can be altered in the counterfactual. We use the fixed point procedure in [Morrow and Skerlos \(2011\)](#) to solve for prices. Solving for equilibrium prices in the flexible model is similar. We solve a modified fixed point in prices described by Equation (D3):

$$\tilde{\nu}_t \tilde{p}_{jt} - \tilde{\tau}_t = \hat{h}(\mathcal{J}(\tilde{p}_t, \tilde{x}_t, \tilde{\xi}_t), D(\tilde{p}_t, \tilde{x}_t, \tilde{\xi}_t), \tilde{w}_{jt}, \tilde{\nu}_t, \tilde{\mathcal{H}}_t) + \tilde{\omega}_{jt}, \quad j = 1, \dots, J, \quad (\text{D3})$$

where $\hat{\omega}_{jt}$ is the residual recovered for product j in market t from the estimated function $\hat{h}$. $\tilde{\omega}_{jt}$ is its potentially altered counterfactual value. To solve the fixed point, we use **df-sane** in the **scipy** package, a derivative-free spectral method with tolerance 1×10^{-6} . Unless otherwise noted below, all variables retain their values without tildes.

Hold-Out Performance After obtaining estimates of our flexible supply model and the parametric supply models, we first compare their performance at predicting prices in a hold-out sample. For each parametric model m , we have estimates $\hat{\gamma}^m$ of the parameters in marginal cost. For observations in the hold-out sample, we recover the implied marginal costs c_{mjt} by inverting the model-implied first-order conditions. We then recover the model-implied unobservable cost shocks from the implied marginal costs as $\hat{\omega}_{mjt} = c_{mjt} - w'_{jt} \hat{\gamma}^m$. For the flexible model of supply, we compute the model-implied unobservable cost shifter by predicting prices from the fitted model and comparing them to observed prices in the hold-out sample. We compute the mean-squared error (MSE) for each of the models; the closer the MSE is to the irreducible error in the true data-generating process (the MSE of the true model), the better the performance.

Computing Cost Pass-through In the baseline environment with constant marginal costs, we also compare the cost pass-through of our estimated flexible supply model to the cost pass-through implied by the true model. In each market, we compute a numerical approximation to the cost pass-through matrices under a given model. Specifically, we obtain

the columns of each pass-through matrix by iterating over the products in the market.²⁶ For each product, we increase its model-implied marginal cost (which corresponds to the true marginal cost under the true model) by 1 percent. We then recompute the equilibrium price vector across all firms in that market. For the true and parametric models, we use the algorithm developed in [Morrow and Skerlos \(2011\)](#) to solve the fixed point in Equation D2. For the estimated flexible model, we perturb the estimated residual $\hat{\omega}_{jt}$ by adding 1 percent of the true marginal cost, denoted $\tilde{\omega}_{jt}$. Under the perturbed $\tilde{\omega}_{jt}$, we solve for the vector of equilibrium prices in market t using Equation D3. Under the parametric and flexible supply models, we report the change in prices divided by the change in cost.

We now explain how we modify markets when computing market counterfactual outcomes. For each counterfactual, we solve for fixed points in prices under a specified parametric model m using Equation D2 and under the flexible model of supply using Equation D3. Unless otherwise noted, all variables remain at their estimation-sample values.

Characteristic Regulation In the first set of market counterfactuals, we evaluate regulations on characteristics that enter the model as cost shifters in the baseline environment. For the parametric model m , we recover costs and perform a regression on observed cost shifters to recover coefficients $\hat{\gamma}^m$ with corresponding residuals $\hat{\omega}^m$. In the case of economies of scale, we additionally regress on observed market shares s_{jt} and squared market shares s_{jt}^2 , instrumenting with product characteristics x_{jt} and squared product characteristics x_{jt}^2 to address endogeneity. For the flexible model, we use the estimated supply function $\hat{h}$ and its corresponding residual.

For this set of counterfactuals, we alter the first indexed cost shifter, $\tilde{w}_{jt}^1 = w_{jt}^1 + 1$. Since w_{jt}^1 is drawn iid $U[0, 1]$, this moves the cost shifter well outside the support in the training data. To evaluate parametric market counterfactuals, we use the estimated $\hat{\gamma}^m$ and $\hat{\omega}^m$ to predict costs with the modified $\tilde{w}_{jt}$:

$$\tilde{c}_{jt}^m = \hat{\gamma}^m \tilde{w}_{jt} + \hat{\omega}_{jt}^m,$$

and solve for equilibrium prices using Equation D2. Similarly, for the flexible model of supply, we input the altered $\tilde{w}_{jt}$ into $\hat{h}$ and solve for equilibrium prices using Equation D3. Under both parametric and flexible supply models, market shares and demand derivatives are updated at each iteration of the fixed point to reflect demand at the current price vector. Under economies of scale, costs can shift with market shares as well. We report differences

²⁶The (j, k) -th element of each pass-through matrix corresponds to the change in price of product j associated with a marginal change in the cost of product k .

in prices for this set of counterfactuals.

Product Characteristic Regulation The next set of counterfactuals is similar to the first, but requires a different procedure to solve the parametric equilibrium. We perform these counterfactuals in the baseline environment. We alter the first indexed product characteristic, $\tilde{x}_{jt}^1 = x_{jt}^1 + 1$. Since x_{jt}^1 is drawn iid $U[0, 1]$, this moves the product characteristic well outside the support in the training data. We then plug in the altered $\tilde{x}_t$ and solve for equilibrium prices using Equations D2 and D3, fixing cost parameters and shocks as above. We report differences in shares for this set of counterfactuals, using the demand system evaluated at the equilibrium prices.

Product Exit Product exit counterfactuals are implemented for the baseline environment. We implement exit by dropping the first firm from each market in which it is present and changing the ownership matrix to $\tilde{\mathcal{H}}$. Under the new ownership matrix, we solve for equilibrium prices using Equations D2 and D3, fixing cost parameters and shocks as above. In the flexible model, we pad the data with zeros to maintain the correct input dimension. Variation in ownership structure helps the neural networks learn the supply function, especially in high-dimensional product spaces. We report differences in consumer surplus across all products and markets using compensating variation as defined in Small and Rosen (1981).

Laffer Curve We implement Laffer curve counterfactuals in the baseline environment with constant marginal cost for two sets of tax instruments: unit and ad valorem taxes. These instruments are implemented in isolation; that is, we consider environments in which only one instrument is implemented at a time. Taxes are assumed to be the same across products. For training, we introduce variation in taxes across markets. Under the unit tax $\tilde{\tau}_t$, we solve for parametric equilibrium prices in Equation D2 by adding $\tilde{\tau}_t$ to the marginal cost. For the flexible model, we load the tax $\tilde{\tau}_t$ onto the residuals such that $\tilde{\omega}_{jt} = \hat{\omega}_{jt} + \tilde{\tau}_t$ and solve for equilibrium prices using Equation D3. Laffer curves are constructed by computing government revenue in a representative market t :

$$G_t^U = \sum_{j \in J_t} \tilde{\tau}_t \tilde{s}_{jt},$$

where $\tilde{s}_{jt}$ are the resulting market shares from the equilibrium prices under unit tax $\tilde{\tau}_t$. Laffer curves are then produced by mapping government revenue G_t^U over the dollar amount of the unit tax $\tilde{\tau}_t$.

For ad valorem taxes, we solve the parametric models by setting counterfactual marginal

costs according to the tax rate:

$$\tilde{c}_{jt}^m = \frac{1}{\tilde{\nu}_t} \tilde{c}_j^m(w_{jt}, \omega_{jt}^m),$$

where $\tilde{\nu}_t = 1/(1 + \tilde{v}_t)$ and $\tilde{v}_t$ is the ad valorem tax rate. For the flexible supply model, we incorporate the tax rate v_t directly as an exogenous variable in training and thus update this value to $\tilde{v}_t$. We then use Equation D3 to solve for equilibrium prices under the counterfactual tax rate. Laffer curves are constructed again by computing government revenue in a representative market t :

$$G_t^A = \sum_{j \in J_t} (1 - \tilde{\nu}_t) \tilde{p}_{jt} \tilde{s}_{jt},$$

where $\tilde{s}_{jt}$ are the resulting market shares from the equilibrium prices under ad valorem tax $\tilde{v}_t$. Laffer curves are then produced by mapping government revenue G_t^A over the ad valorem tax rate $\tilde{v}_t$.

Merger Simulation Merger simulation counterfactuals are implemented for the merger-simulation environment. We implement the merger by unifying the firm identifiers for the two merging firms (i.e., the two three-product firms) and changing the ownership matrix to $\tilde{\mathcal{H}}_t$. Under the new ownership matrix, we solve for equilibrium prices using Equations D2 and D3, fixing cost parameters and shocks as above. In the flexible model, we pad the data with zeros to maintain the correct input dimension. Variation in ownership structure helps the neural networks learn the supply function, especially in this high-dimensional environment. By construction, the ownership structure in markets affected by the merger is “in-sample” in the sense that this exact market structure was observed during training. We report differences in consumer surplus across all products and markets using compensating variation as defined in Small and Rosen (1981).

D.5 Quantifying Uncertainty

Quantification of uncertainty also requires the specification of hyperparameters because it involves training another deep neural network. As in estimation, we use OAdam, setting the learning rate to $\eta = 5 \times 10^{-2}$ and the momentum decay parameters to $\beta = [0.5, 0.9]$. We include all markets in a single batch during gradient descent and implement a smooth start by averaging over the last 4,000 epochs of training. The dimensionality of f mirrors that of estimation with 50 nodes in the first hidden layer and 20 nodes in the second. We omit regularization terms. The network f is fully connected with leaky ReLU activation functions. In computing $\nabla_{\theta} \omega(\theta_0)$, we use automatic differentiation in `torch` to differentiate

the loss function with respect to the parameters of the deep neural network, stacking the resulting derivatives into a $b \times 1$ vector consistent with the variance objective function. In Algorithm 1, we take T_α from a folded normal distribution with tuning parameter $c = 1$ and $\alpha = 0.05$ as our fixed critical values. In the figures throughout, we use the Bonferroni correction as a more conservative approach for the sake of interpretability; our inference algorithm generates strictly tighter confidence intervals.

E Additional Empirical Results

E.1 Data Construction

We construct a database of the U.S. airline industry from 2005–2019. We obtained a quarterly random 10 percent sample of purchased airline tickets from the well-known Airline Origin and Destination Survey (DB1B) database released by the U.S. Department of Transportation. Following Azar, Schmalz, and Tecu (2018) and Kennedy, O’Brien, Song, and Waehrer (2017), a market is defined as a pair of cities, regardless of the flight direction. We match cities to Metropolitan Statistical Areas and collect data on the populations of these MSAs from the Bureau of Economic Analysis. A product is a one-way trip that services a particular city-pair and is defined at the carrier-market-quarter level. Market sizes are measured as the geometric mean of the origin-destination endpoint populations.

E.1.1 Sample Selection

We exclude markets with fewer than 20 passengers per day, as airline behavior on these thin, possibly seasonal, routes is unlikely to represent normal competitive behavior in the industry. We also drop itineraries with a ticket carrier change at the connecting airport since these tickets cannot be assigned to a unique ticketing carrier. Finally, we drop every ticket with a fare lower than \$25 and higher than \$2,500 since these tickets are likely the result of reporting errors.

For each carrier-market-quarter, we begin by calculating the product’s average price, total passengers, and average distance. Additionally, we construct each product’s extra miles, defined as the difference between average distance in miles and nonstop distance in the market, and the fraction of nonstop tickets sold. Averages are weighted by the number of passengers. We remove any products with fewer than 800 quarterly passengers. This restriction is standard in this literature (e.g., Berry and Jia, 2010). It is especially useful in our application, where the dimensionality of the input space matters, because it allows us

to retain more markets with effectively fewer carriers. Summary statistics for the analysis sample are presented in Table E1.

TABLE E1: Summary Statistics

Statistic	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
Average Fare	216.8	82.8	25.0	169.8	209.1	252.7	2,492.0
Total Passengers	352.4	1,184.0	1	19	69	210	70,909
Average Distance	1,386.5	688.7	67.0	861.0	1,255.1	1,872.0	7,731.5
Average Nonstop Miles	1,171.0	619.0	67.0	678.0	1,035.0	1,600.0	2,783.0
Average Extra Miles	215.6	247.4	-1.0	37.5	136.0	305.9	5,118.0
Share Nonstop	0.2	0.4	0.0	0.0	0.0	0.2	1.0
Origin Hub	0.2	0.4	0	0	0	0	1
Dest. Vacation	0.1	0.3	0	0	0	0	1
LCC	0.2	0.4	0	0	0	0	1
Major	0.9	0.3	0	1	1	1	1
Legacy	0.7	0.4	0	0	1	1	1
Presence	0.1	0.1	0.0	0.1	0.1	0.2	1.0
Num Markets	50.4	30.2	1	27	47	72	146
Share (%)	0.1	0.3	0.0	0.0	0.0	0.1	9.5
Within Share	0.2	0.2	0.0	0.0	0.1	0.3	1.0

Notes: Summary statistics for the analysis sample.

E.2 Demand Estimation

We include additional details on demand estimation introduced in Section 6.2. In our demand system, consumer i receives utility from product j in market t with the following indirect utility:

$$u_{ijt} = \alpha_p p_{jt} + x_{jt} \beta + \xi_{m(t)} + \xi_{jt} + \zeta_{it} + (1 - \rho) \varepsilon_{ijt}$$

The vector x_{jt} includes the share of nonstop flights, average distance in thousands of miles, the squared term of average distance in thousands of miles, and the logged number of fringe firms (plus one to avoid zero issues). The last term is included to focus on the demand for major carriers while controlling for additional variation over time in market structure across origin-destination pairs. The term $\xi_{m(t)}$ is a set of origin-destination fixed effects. ξ_{jt} and $\zeta_{it} + \varepsilon_{ijt}$ are unobservable shocks at the product-market and individual-product-market

levels, respectively. We assume that ε_{ijt} is distributed Type I Extreme Value and ζ_{it} is distributed according to the conjugate distribution (Cardell, 1997). We close the model by normalizing the utility of consumer i from the outside option to $u_{i0t} = \varepsilon_{i0t}$. Given the structure of utility and distributional assumptions, market shares s_{jt} are a function of observables, unobservables, and parameters in the standard form from Berry et al. (1995).

The identifying assumption for demand is that the moment condition $\mathbb{E}[\xi_{jt}z_{jt}^D] = 0$ holds for a vector of demand instruments z_{jt}^D . Following Berry et al. (1995), we include the average rival distance, the average number of markets a rival serves, and the number of rival carriers. The last instrument is especially useful for identifying the nesting parameter. The results of demand estimation are presented in Table E2. The results and median own-price elasticities are in line with the literature.

TABLE E2: Demand Estimates

	$\log(s_{jt}) - \log(s_{0t})$
Average Fare	-0.0048*** (0.0004)
$\log(S_t)$	0.8356*** (0.0133)
Share Nonstop	0.4030*** (0.0282)
Average Distance (1,000's)	-0.4881*** (0.0498)
Average Distance ² (1,000's)	0.0485*** (0.0045)
$\log(1 + \text{Num. Fringe})$	-0.2642*** (0.0057)
R ²	0.94238
Observations	1,283,472
Own-price elasticity	-5.1652
Origin-destination fixed effects	✓

Notes: Prices and $\log(S_t)$ instrumented with average rival distance, average number of rival markets, and number of rival carriers. Origin-destination fixed effects; standard errors clustered at the origin-destination level.

E.3 Supply Estimation

In Section 6.2, we briefly introduced the supply side and focused on the flexible markup function. In the Bertrand specification, we assume that inferred marginal costs are linear in observable cost shifters:

$$p_{jt} - \Delta_{jt}^B \equiv c_{jt} = w_{jt}\gamma + \Gamma_{m(t)} + \omega_{jt}$$

We include only the average distance in thousands of miles in w_{jt} and origin-destination

fixed effects as $\Gamma_{m(t)}$. In a second specification, we introduce economies of scale by including and instrumenting for market shares. The instruments are product characteristics excluded from supply. The results are presented in Table E3. We find that distance is positively associated with marginal costs inferred under the Bertrand assumption of conduct. The second specification indicates economies of scale, with a large, negative, and significant coefficient on market shares.

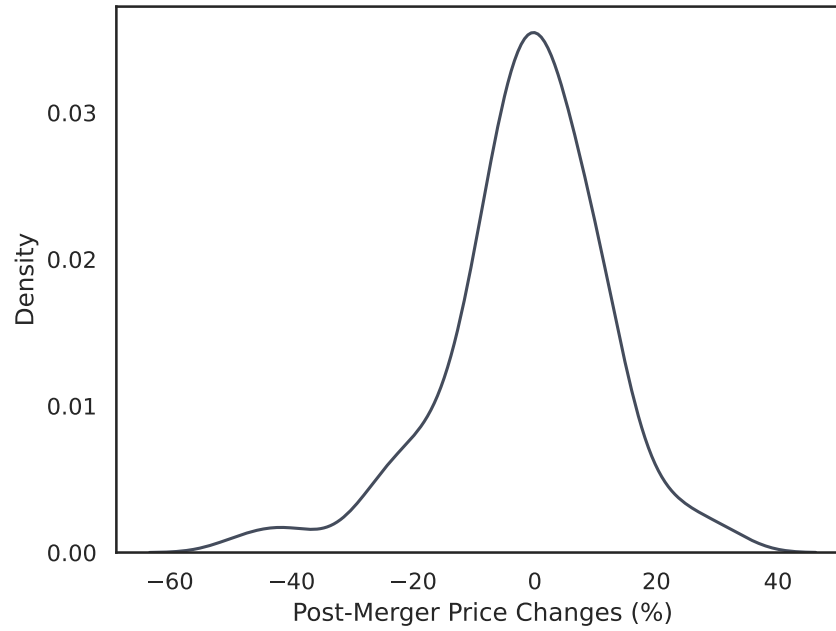
TABLE E3: Bertrand-Implied Marginal Cost Estimates

	Marginal Cost	
Average Distance (1,000's)	63.17*** (0.9502)	28.05*** (2.821)
Market Share (%)		-113.5*** (10.23)
R ²	0.42757	0.39670
Observations	1,283,472	1,283,472
Origin-destination fixed effects	✓	✓

Notes: Origin-destination fixed effects; standard errors clustered at the origin-destination level.

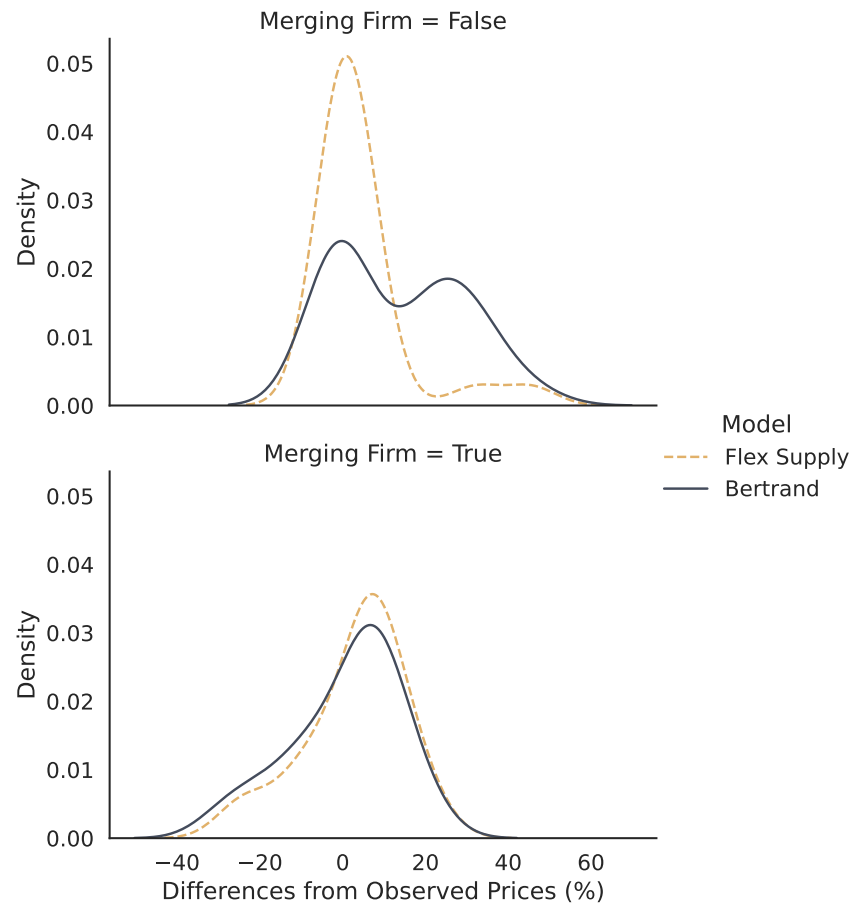
E.4 Counterfactuals

FIGURE E1: Price Change Distribution



Notes: The figure plots the observed post-merger price changes after the American Airlines-US Airways merger.

FIGURE E2: Post-Merger Price Prediction Error by Status



Notes: Merger simulation results broken down by firm status: flexible model (yellow) versus Bertrand (blue). Distribution of percent differences between observed and predicted post-merger prices.

F Using VMM

Researchers can easily use our implementation of VMM in simulations and empirical applications. The code is available on GitHub at github.com/jackcollison/pyvmm. The models are implemented using the `torch` package in Python. Compute times are manageable by modern standards. Our simulations for $T = 100$ finish within a few minutes; $T = 1,000$ within an hour; and $T = 10,000$ in about a day. The model for our empirical application runs in about 16 hours. We build an easy-to-use wrapper around the machinery of VMM. Researchers specify cost shifters, product characteristics, and instruments they would like to include. We allow the user to optionally include demand derivatives, which we build in the background.

Online Appendix References

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Online Supplemental Materials

This document collects supplemental materials for *Market Counterfactuals with Nonparametric Supply: An ML/AI Approach*. It accompanies the Online Appendix above. Section S.1 develops cost and markup decompositions through model extensions. Section S.2 details the data construction underlying the empirical application. Section S.3 reports additional simulation and counterfactual results.

S.1 Extensions: Decomposing Costs and Markups, and Other Economic Restrictions

While our main identification result in Section 3.2 establishes nonparametric identification of the supply function $h_j(s_t, D_t, w_{jt})$, certain counterfactual exercises require separate identification of the cost and markup components. For instance, evaluating technological improvements requires knowledge of the cost function, while assessing changes in competitive conduct requires identification of markups. We outline two approaches to achieve this decomposition, though we do not pursue these extensions because they lie outside our primary scope.

S.1.1 Separating Markup and Cost Through Market Size Variation

The first approach leverages exogenous variation in market size M_t to separately identify cost and markup functions without imposing parametric restrictions on either component.

Recall that quantities are $q_{jt} = M_t s_{jt}$. When market size varies, holding market shares fixed, costs change through the quantity channel while markups remain constant because they depend only on shares and demand derivatives. Under the decomposition:

$$h_j(s_t, D_t, w_{jt}) = c_j(M_t s_{jt}, w_{jt}) + \Delta_j(s_t, D_t),$$

the key insight is that for fixed (s, D, w) , variation in M_t affects only the cost component. Taking the derivative with respect to M :

$$\frac{\partial}{\partial M} \mathbb{E} [p_{jt} \mid s_t = s, D_t = D, w_{jt} = w, M_t = M] = \frac{\partial c_j}{\partial q_{jt}} \cdot s.$$

This identifies the marginal cost at each quantity level. Integrating from a baseline quantity

q_0 :

$$c_j(q, w) - c_j(q_0, w) = \int_{q_0}^q \frac{1}{s} \frac{\partial}{\partial M} \mathbb{E} [p_{jt} \mid s_t = s, D_t = D, w_{jt} = w, M_t = \tilde{q}/s] d\tilde{q}.$$

With a normalization for $c_j(q_0, w)$, the cost function is identified. The markup function follows as:

$$\Delta_j(s, D) = \mathbb{E} [p_{jt} \mid s_t = s, D_t = D, w_{jt} = w, M_t = M] - c_j(Ms, w).$$

This approach requires substantially stronger data requirements than our baseline identification. Most importantly, it requires observing the same market configuration (s_t, D_t) across different market sizes, essentially requiring that M_t varies independently of equilibrium shares and demand derivatives. This may be violated if market size itself affects equilibrium outcomes through entry, product positioning, or competitive intensity. Moreover, the completeness condition must be augmented to ensure that variation in M_t provides sufficient information to separately identify both cost and markup functions. In many empirical settings, such rich variation in market size may be unavailable or confounded with other market characteristics. More broadly, even when M_t variation is available, constructing instruments that isolate the cost channel while controlling for the simultaneous determination of shares and derivatives remains a practical challenge, as the required conditional independence between M_t and market-level unobservables may be difficult to justify.

S.1.2 Separating Markup and Cost Through Economic Restrictions

The second approach imposes structure on either conduct or costs to achieve separate identification. We consider each possibility in turn.

S.1.2.1 Known Conduct

Following [Berry and Haile \(2014\)](#) Section 4.3, suppose the form of oligopoly competition is known:

Assumption 10 (Known Oligopoly Model). Markups take the form $\Delta_j(s_t, D_t) = \psi_j(s_t, D_t)$ where ψ_j are known functions determined by the oligopoly model.

For example, under Bertrand competition with multi-product firms, ψ_j is the (j, j) -th element of the matrix Γ_t^{-1} where the (j, k) -th element of Γ_t is equal to $\partial s_k / \partial p_j$ when products j and k are produced by the same firm, and zero otherwise.

Under Assumption 10, marginal costs are directly identified:

$$c_j(q_{jt}, w_{jt}) + \omega_{jt} = p_{jt} - \psi_j(s_t, D_t).$$

The cost function $c_j(\cdot, \cdot)$ can then be identified nonparametrically using instruments for quantity, following the arguments in [Berry and Haile \(2014\)](#) Theorem 6. This requires that demand shifters x_{jt} are excluded from marginal costs, allowing them to serve as instruments for the endogenous quantity in the cost function regression.

The key advantage of this approach is that it requires only standard instrumental variables variation rather than the stringent market size conditions of Section [S.1.1](#). The disadvantage is the strong assumption of known conduct, which rules out testing between alternative models of competition.

S.1.2.2 Parametric Costs

Alternatively, we can impose a functional form for costs while maintaining flexibility in conduct:

Assumption 11 (Parametric Cost Structure). Marginal costs take the parametric form $c_j(q_{jt}, w_{jt}; \theta_j) = c(w_{jt}, q_{jt}; \theta_j)$ for a known function $c(\cdot, \cdot; \theta_j)$ with finite-dimensional parameters θ_j .

A common specification is log-linear costs: $c_j(q_{jt}, w_{jt}; \theta_j) = w'_{jt}\gamma_j + \alpha_j \log(q_{jt})$ where $\theta_j = (\gamma_j, \alpha_j)$. This transforms our nonparametric identification problem into a semi-parametric one. The supply equation becomes:

$$p_{jt} = c(w_{jt}, M_t s_{jt}; \theta_j) + \Delta_j(s_t, D_t) + \omega_{jt}.$$

The parameters θ_j can be estimated in a first stage using the orthogonality condition $\mathbb{E}[\omega_{jt}|z_t, w_t] = 0$ and excluded instruments. Given consistent estimates $\hat{\theta}_j$, the markup function is identified nonparametrically as:

$$\Delta_j(s_t, D_t) = h_j(s_t, D_t, w_{jt}) - c(w_{jt}, M_t s_{jt}; \hat{\theta}_j)$$

This semi-parametric approach represents a middle ground: it imposes less structure than assuming known conduct while requiring weaker data requirements than the fully nonparametric approach of Section [S.1.1](#). The parametric restrictions on costs are often more palatable than conduct assumptions, as they can be motivated by production theory or tested against more flexible specifications.

S.1.3 Implementing Economic and Statistical Restrictions

As discussed above, one may want to impose additional economic or statistical restrictions on the h function. In addition to the separability between cost and markups, one may want, e.g., h_j to be decreasing in that product's own demand elasticity, as is the case in many standard models. This is in the spirit of the micro-founded economic restrictions that are imposed on nonparametric demand systems in [Compiani \(2022\)](#) and [Brand and Smith \(2025\)](#); such restrictions can also be formally tested (see [Breunig and Chen, 2024](#), for adaptive tests of shape restrictions in nonparametric IV models). It is possible to add these restrictions to our model through regularization, restrictions on weights and activation functions, neural network architecture, or some combination of these. We discuss each of these in turn within the example of monotonicity in own demand elasticities.

The first approach is to incorporate additional components in the regularization term R_N in Equation (4). Define a set of n own-demand elasticities in increasing order as $\eta = (\eta_{(1)}, \dots, \eta_{(n)})$. An example of a simple regularization term R^M for monotonicity is the following:

$$R^M(h) = \sum_{i=2}^n (\max\{h(\eta_{(i)}) - h(\eta_{(i-1)}), 0\})^2$$

We condition on s_t , w_t , θ , and $\mathcal{H}_t$ in the markup function h , suppressing them for notational simplicity. If $h(\eta_{(i)}) \leq h(\eta_{(i-1)})$, there is no additional penalty on the markup function, but otherwise, we penalize the squared first difference of own-demand elasticities in the spirit of a ridge regression. We note that the choice of regularization is a degree of freedom for the researcher.

The last two approaches are related to the rich computer science literature on monotonic neural networks, starting with [Sill \(1997\)](#). The first approach enforces constraints on weights and activation functions in particular neural network layers, e.g., [You, Ding, Canini, Pfeifer, and Gupta \(2017\)](#). The user can specify inputs, such as own-demand elasticities, in which the output is monotonic. A second approach, detailed in [Wehenkel and Louppe \(2019\)](#), uses the architecture of the neural network to enforce a constant sign of the derivative of the approximated function without imposing additional constraints. In our specific example, we can restrict the derivative of h with respect to own-demand elasticity to be negative.

The discussion above focuses on a single example of an economic restriction. These approaches can be adapted and combined to impose additional economic or statistical restrictions on the markup function.

S.2 Data Construction for Ownership-Based Ordering

This section details the implementation of the ownership-based ordering described in Assumption 6. We show how market shares, demand derivatives, and instruments are systematically reordered to embed ownership structure into the supply function estimation.

S.2.1 General Construction Procedure

Consider a market t with products indexed by j and firms indexed by f . For each product j owned by firm f , we construct the following ordered vectors:

Outcomes: The dependent variable is simply the price p_{jt} .

Exogenous Variables: Own cost shifters w_{jt} enter directly without reordering.

Market Shares: We reorder market shares using the ownership matrix. First, we place the own market share s_{jt} . Second, we include other products owned by the same firm, denoted $s_{-j,f,t}$. If firm f owns fewer than the maximum number of products $\bar{J}_f$ observed across all firms in the sample, we pad with zeros to maintain consistent dimensions. Third, we append rival firms' market shares $s_{-f,t}$.

Demand Derivatives: The demand derivative matrix is partitioned into blocks following the ownership structure. We order own-price elasticities as $(D_{jj,t}, D_{(-j,f),(-j,f),t}, D_{-f,-f,t})$, padding with zeros as needed. Cross-price elasticities are ordered as own-firm cross-elasticities $D_{j,(-j,f),t}$ followed by cross-firm elasticities $D_{j,-f,t}$.

Instruments: Product characteristics and rival cost shifters are ordered following the same pattern as market shares: $(x_{jt}, x_{-j,f,t}, x_{-f,t}, w_{-f,t})$.

S.2.2 Example: Duopoly with Multi-Product Firm

Consider a market with a single-product firm (Firm 0) and a two-product firm (Firm 1). The raw data contains:

TABLE S.2.1: Raw Product Data

Firm	Product	x_1	w_1	s	p
0	0	0.4	0.1	0.4	2
1	1	0.2	0.2	0.2	3
1	2	0.3	0.3	0.3	4

Under logit demand with price coefficient $\alpha_p = -1$, the demand derivative matrix is:

$$D_t = \begin{bmatrix} -0.24 & 0.08 & 0.12 \\ 0.08 & -0.16 & 0.06 \\ 0.12 & 0.06 & -0.21 \end{bmatrix}$$

After applying the ownership-based ordering, the constructed data for estimation becomes:

TABLE S.2.2: Reordered Market Shares and Cost Shifters

Firm	Product	w_1	Own Firm		Rival Firm		p
			s_j	$s_{-j,f}$	$s_{-f,0}$	$s_{-f,1}$	
0	0	0.1	0.4	0	0.2	0.3	2
1	1	0.2	0.2	0.3	0.4	0	3
1	2	0.3	0.3	0.2	0.4	0	4

Note that Product 0, being the only product of Firm 0, has $s_{-j,f} = 0$ (padded). Products 1 and 2, both owned by Firm 1, include each other's shares in the $s_{-j,f}$ column and pad the second rival column with zeros since only one rival firm exists.

The demand derivatives are similarly reordered:

TABLE S.2.3: Reordered Own-Price Demand Derivatives

Firm	Prod.	Own Elasticities		Rival Own-Elasticities	
		D_{jj}	$D_{(-j,-j),f}$	$D_{-f0,-f0}$	$D_{-f1,-f1}$
0	0	-0.24	0	-0.16	-0.21
1	1	-0.16	-0.21	-0.24	0
1	2	-0.21	-0.16	-0.24	0

TABLE S.2.4: Reordered Cross-Price Demand Derivatives

Firm	Prod.	j Own Cross	j Rival Cross		$-j$ Own Cross	$-j$ Rival Cross
		$D_{j,f}$	$D_{j,-f0}$	$D_{j,-f1}$	$D_{-j,f}$	$D_{(-j,-f),-f}$
0	0	0	0.08	0.12	0	0.06
1	1	0.06	0.08	0	0.12	0
1	2	0.06	0.12	0	0.08	0

This systematic reordering allows the neural network to learn patterns that depend on ownership structure without explicitly including the ownership matrix as a separate input, effectively embedding the economic structure of multi-product firm decision-making into the functional form of the estimator.

S.3 Additional Simulation and Counterfactual Results

S.3.1 Additional Hold-Out Sample Results

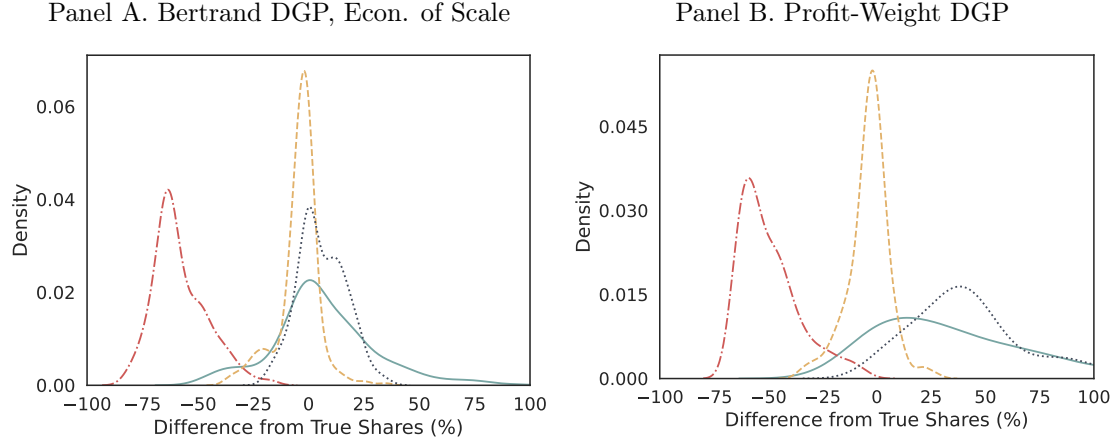
TABLE S.3.5: MSE Ratios Across Models, Baseline Environment with Constant Costs

T	Standard Models			Flexible Models			TF-RF	D_t
	B	M	P	$h = 3$	$h = 20$	$h = 100$		
Panel A: Bertrand DGP								
100	0.98	1939.92	6.33	[1.59, 1.64]	[1.20, 1.61]	[1.18, 1.31]	[6.41, 13.00]	No
				[1.09, 1.60]	[1.06, 1.11]	[1.15, 1.23]		Yes
1,000	1.00	1655.16	10.28	[2.31, 2.38]	[1.02, 1.06]	[1.06, 1.14]	[5.96, 7.59]	No
				[1.22, 1.24]	[1.01, 1.09]	[1.02, 1.07]		Yes
10,000	1.00	3023.84	9.88	[2.72, 2.77]	[0.98, 1.00]	[0.99, 1.00]	[6.07, 6.66]	No
				[1.24, 1.25]	[0.98, 1.00]	[0.98, 0.99]		Yes
Panel B: Profit-Weight DGP								
100	4.47	103.73	7.41	[1.62, 1.70]	[1.35, 1.61]	[1.44, 1.63]	[6.74, 14.92]	No
				[1.34, 1.53]	[1.36, 1.54]	[1.34, 1.60]		Yes
1,000	4.54	68.10	10.23	[2.78, 2.81]	[1.10, 1.31]	[1.05, 1.10]	[6.29, 7.94]	No
				[1.26, 1.29]	[0.98, 1.01]	[1.00, 1.03]		Yes
10,000	4.56	119.26	9.95	[2.98, 3.03]	[1.04, 1.13]	[1.02, 1.03]	[6.24, 6.72]	No
				[1.26, 1.27]	[0.98, 1.01]	[0.99, 1.00]		Yes

Notes: MSE ratios relative to the correctly specified model (ratio of 1 = equal performance). B = Bertrand, M = monopoly, P = perfect competition, all with constant cost. TF-RF = reduced-form transformer trained on cost shifters and product characteristics. MSE computed on a hold-out test sample of markets not used in training. Brackets are the 10th and 90th percentiles across 50 simulation runs; see Section 5.

S.3.2 Additional Counterfactual Results

FIGURE S.3.1: Regulation of Product Characteristics: Share Predictions

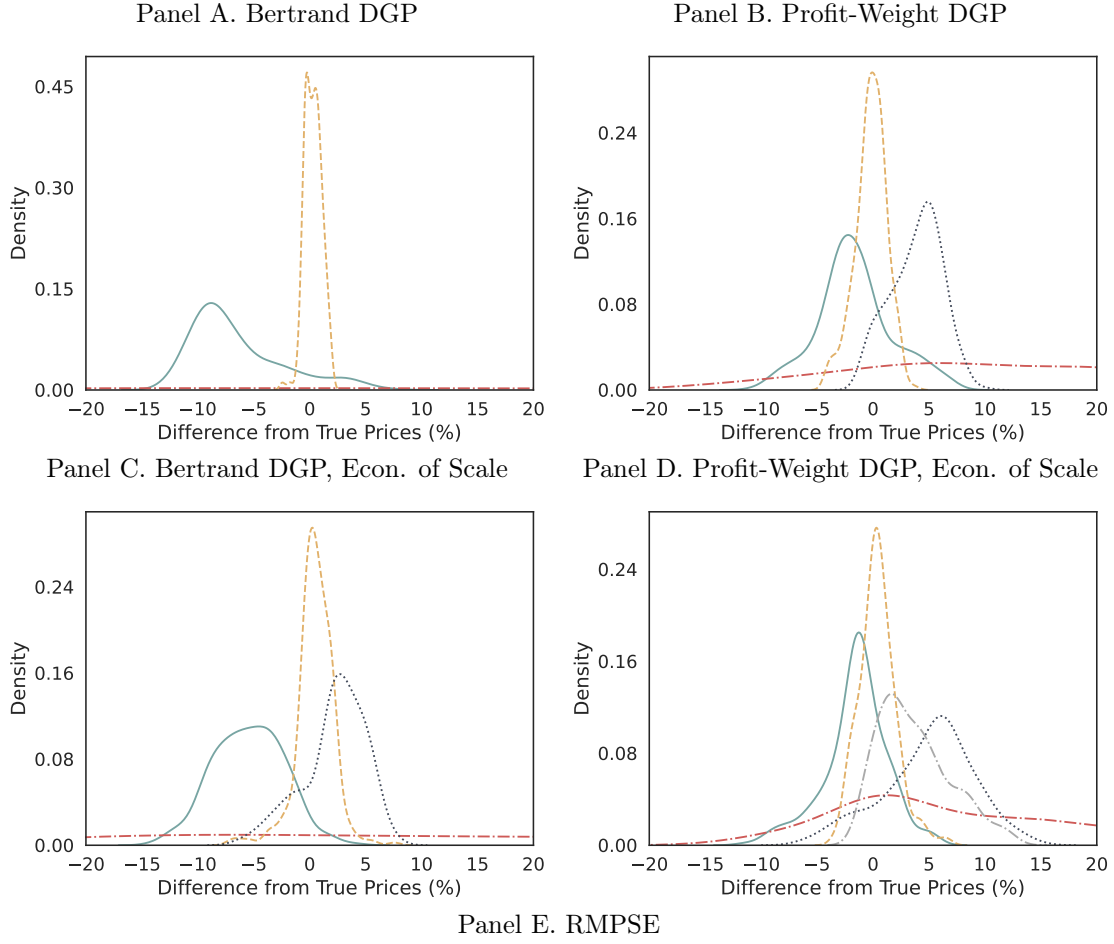


Panel C. RMPSE

Fitted Model	DGP			
	Bertrand	Profit-Weight	Bertrand (Scale)	Profit-Weight (Scale)
--- Bertrand (Scale)	-	-	-	[77.45, 87.40]
.... Bertrand (Const.)	-	[54.04, 56.41]	[12.92, 14.22]	[97.56, 106.26]
--- Monopoly	[58.37, 58.86]	[51.09, 51.36]	[58.28, 59.08]	[42.89, 44.15]
— Perf. Comp.	[23.31, 24.78]	[60.64, 61.96]	[25.35, 26.62]	[77.46, 79.97]
--- Flex Supply	[3.70, 5.28]	[9.01, 11.64]	[9.53, 11.43]	[17.15, 23.57]
TF-RF	[42.57, 48.77]	[38.43, 43.94]	[41.47, 47.14]	[37.23, 42.72]

Notes: Share prediction errors when $\tilde{x}_1 = x_1 + 1$ shifts out of training support. Flexible model: $|h| = 20$, D_t included, $T = 1,000$. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

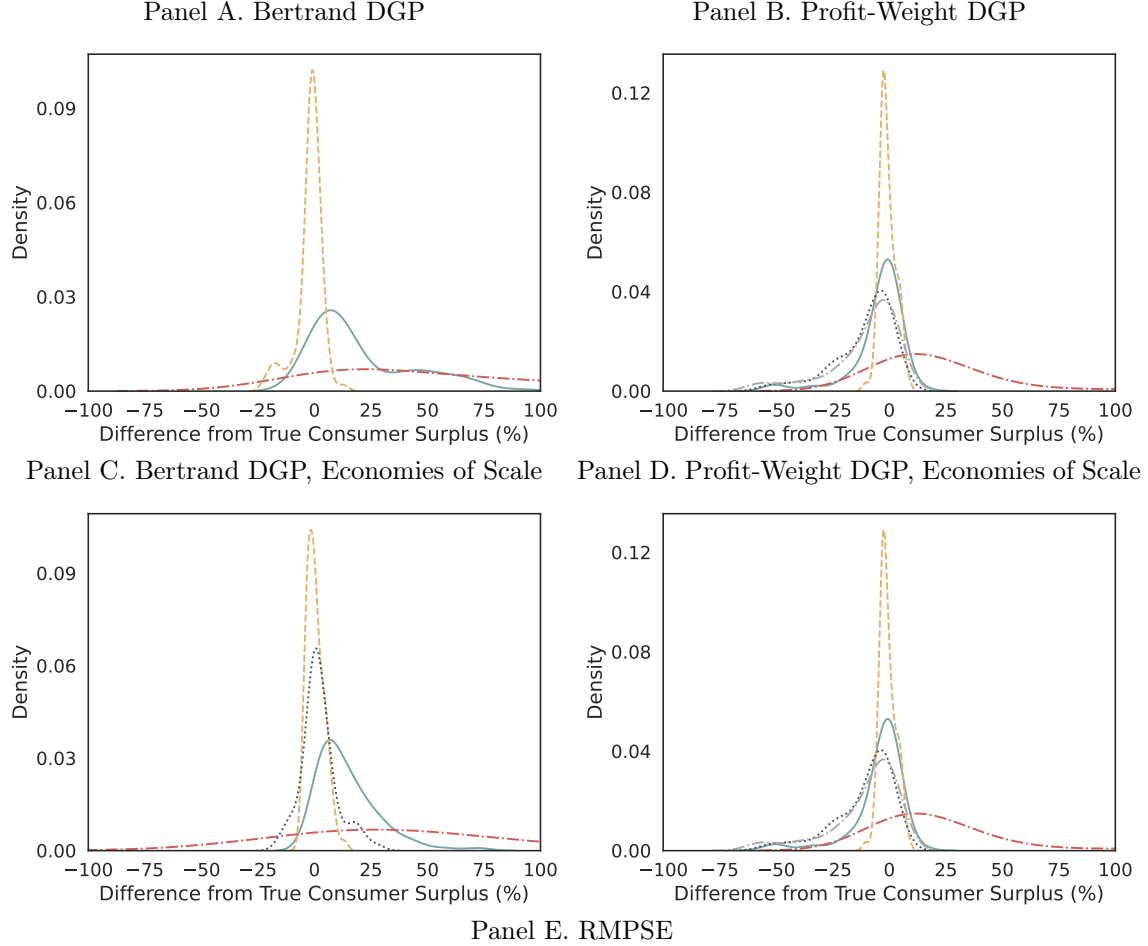
FIGURE S.3.2: Regulation of Cost Shifters: Price Predictions



Fitted Model	Panel DGP			
	Bertrand	Profit-Weight	Bertrand (Scale)	Profit-Weight (Scale)
--- Bertrand (Scale)	-	-	-	[5.16, 6.46]
..... Bertrand (Const.)	-	[4.52, 5.95]	[2.72, 3.62]	[6.92, 8.46]
- - - Monopoly	[92.78, 144.86]	[11.54, 17.54]	[48.30, 74.65]	[8.52, 11.03]
— Perf. Comp.	[6.82, 8.83]	[3.28, 4.39]	[5.71, 7.55]	[2.81, 3.75]
- - - Flex Supply	[0.93, 1.86]	[1.35, 1.66]	[1.39, 1.89]	[2.35, 3.89]

Notes: Price prediction errors when w_1 is doubled. Flexible model: $|h| = 20$, D_t included, $T = 1,000$. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

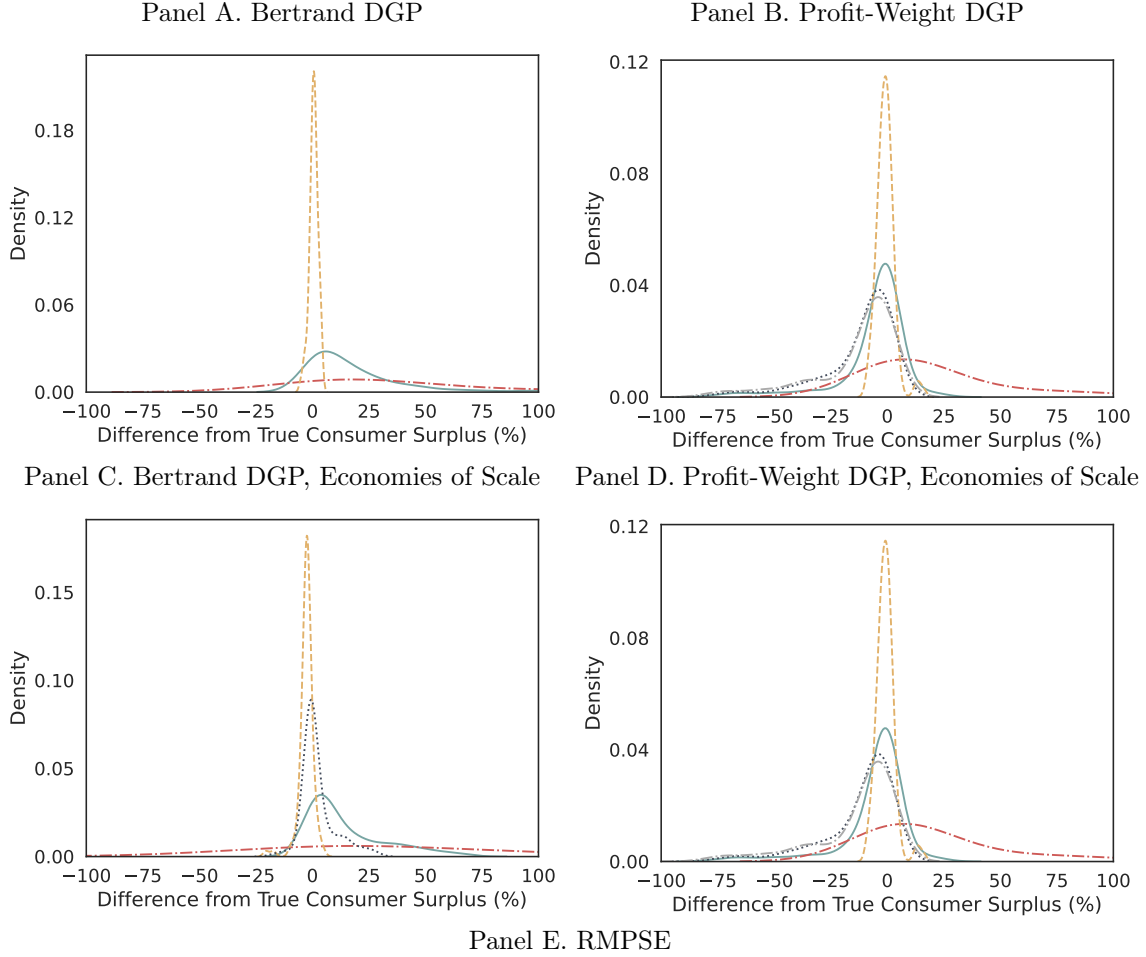
FIGURE S.3.3: Single-Product Merger Simulation



Fitted Model	Panel DGP			
	Bertrand	Profit-Weight	Bertrand (Scale)	Profit-Weight (Scale)
--- Bertrand (scale)	-	-	-	[20.43, 22.55]
..... Bertrand (Const.)	-	[22.59, 24.08]	[7.46, 8.63]	[19.31, 21.03]
- - - Monopoly	[107.90, 135.69]	[43.35, 50.12]	[131.03, 172.88]	[58.90, 71.75]
— Perf. Comp.	[27.64, 29.86]	[19.79, 21.55]	[20.13, 21.62]	[15.04, 17.19]
- - - Flex Supply	[2.84, 4.41]	[2.88, 4.48]	[6.20, 8.52]	[5.37, 8.96]

Notes: Distribution and MSE of merger price predictions. Flexible model: $|h| = 3$, $T = 1,000$; profit-weight panels include D_t . Trained on duopolies and triopolies. MSE computed on a hold-out test sample restricted to markets with merging firms. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

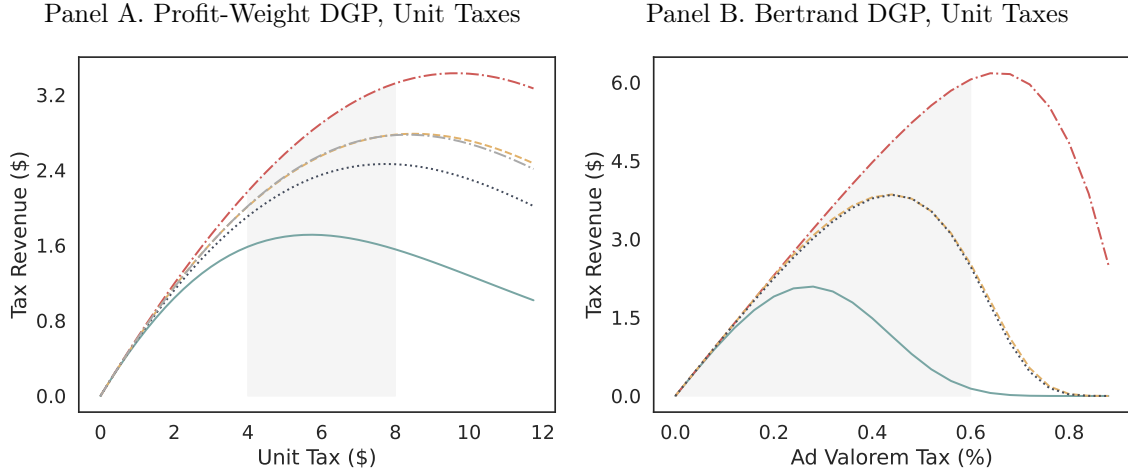
FIGURE S.3.4: Single-Product Merger Simulation (Triopolies)



Fitted Model	Panel DGP			
	Bertrand	Profit-Weight	Bertrand (Scale)	Profit-Weight (Scale)
--- Bertrand (Scale)	-	-	-	[22.80, 24.60]
..... Bertrand (Const.)	-	[22.37, 23.64]	[7.77, 8.77]	[20.26, 21.79]
--- Monopoly	[88.64, 96.88]	[40.24, 42.72]	[119.50, 131.66]	[59.68, 64.05]
— Perf. Comp.	[23.99, 26.11]	[19.56, 21.36]	[21.95, 23.54]	[16.25, 17.94]
--- Flex Supply	[2.51, 3.83]	[2.25, 3.49]	[4.67, 6.77]	[4.58, 6.49]

Notes: Distribution and MSE of merger price predictions. Flexible model: $|h| = 3$, $T = 1,000$; profit-weight panels include D_t . Trained on triopolies. MSE computed on a hold-out test sample restricted to markets with merging firms. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

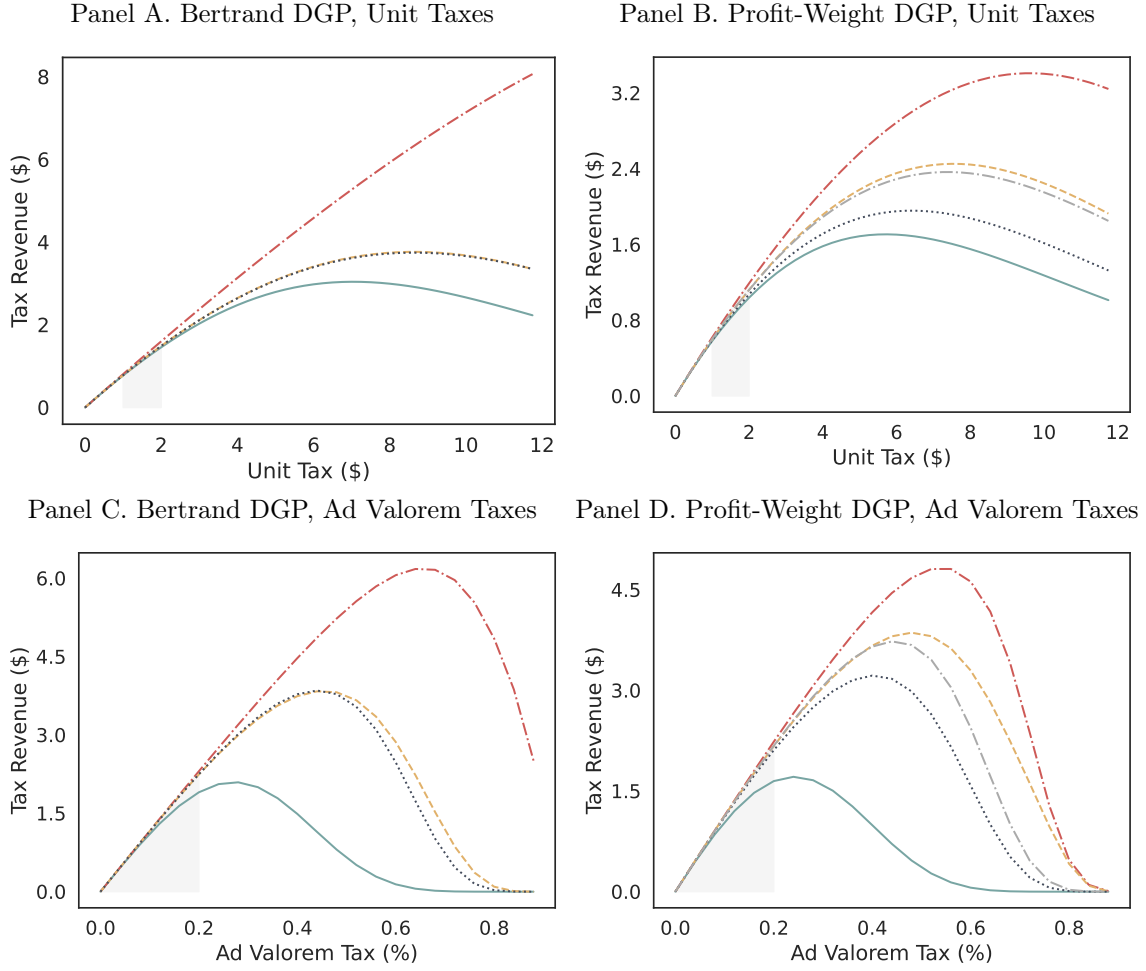
FIGURE S.3.5: Laffer Curves for Unit and Ad Valorem Taxes



Fitted Model		DGP			
		Bertrand (U)	Profit-Weight (U)	Bertrand (AV)	Profit-Weight (AV)
.....	Bertrand	-	[0.21, 0.37]	-	[0.36, 0.65]
---	Monopoly	[1.03, 1.85]	[0.42, 0.79]	[2.09, 3.28]	[0.87, 2.10]
—	Perf. Comp.	[0.34, 0.71]	[0.56, 0.98]	[1.03, 1.73]	[1.19, 1.92]
- - -	Flex Supply	[0.01, 0.04]	[0.02, 0.03]	[0.03, 0.08]	[0.06, 0.14]
	TF-RF	[1.19, 3.29]	[1.07, 3.01]	[1.70, 7.09]	[1.50, 7.26]

Notes: Laffer curves under different conduct assumptions. Training support: unit taxes $\sim U[4, 8]$, ad valorem $\sim U[0, 0.6]$. Flexible model: $|h| = 20$, D_t included, $T = 1,000$. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

FIGURE S.3.6: Laffer Curves for Unit and Ad Valorem Taxes



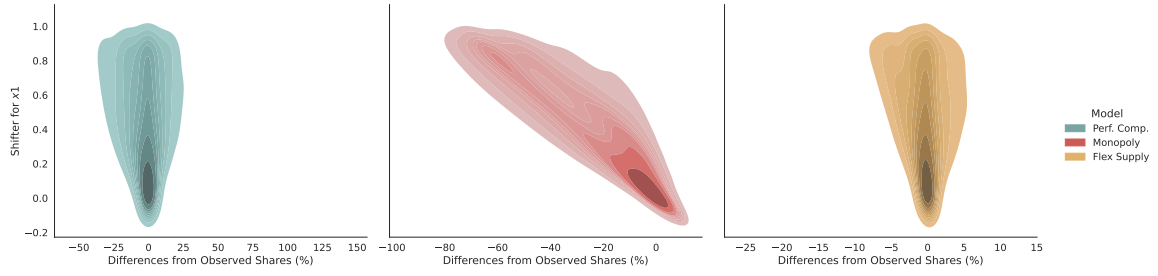
Fitted Model

Panel DGP

		Bertrand (U)	Profit-Weight (U)	Bertrand (AV)	Profit-Weight (AV)
.....	Bertrand	-	[0.21, 0.37]	-	[0.36, 0.65]
---	Monopoly	[1.03, 1.85]	[0.42, 0.79]	[2.09, 3.28]	[0.87, 2.10]
—	Perf. Comp.	[0.34, 0.71]	[0.56, 0.98]	[1.03, 1.73]	[1.19, 1.92]
- - -	Flex Supply	[0.03, 0.09]	[0.02, 0.08]	[0.14, 0.34]	[0.19, 0.56]

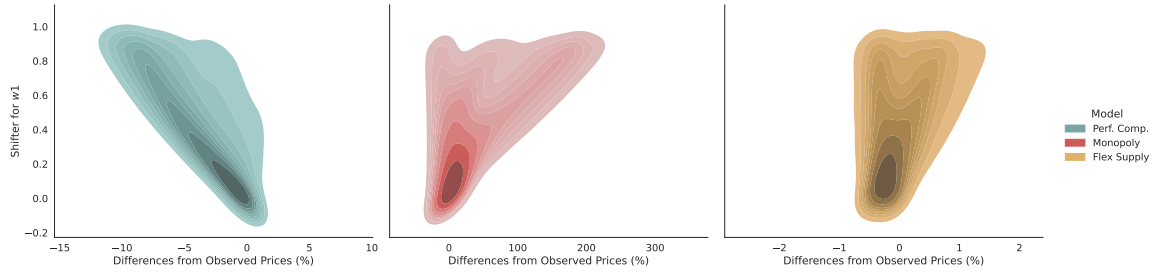
Notes: Laffer curves with reduced tax variation: unit taxes $\sim U[1, 2]$, ad valorem $\sim U[0, 0.2]$. Flexible model: $|h| = 20$, D_t included, $T = 1,000$. Brackets are the 10th and 90th percentiles across 50 simulation runs. The plot panels show the flexible model with the median mean-squared error across the 50 runs.

FIGURE S.3.7: Product Characteristics Regulation, Bertrand DGP



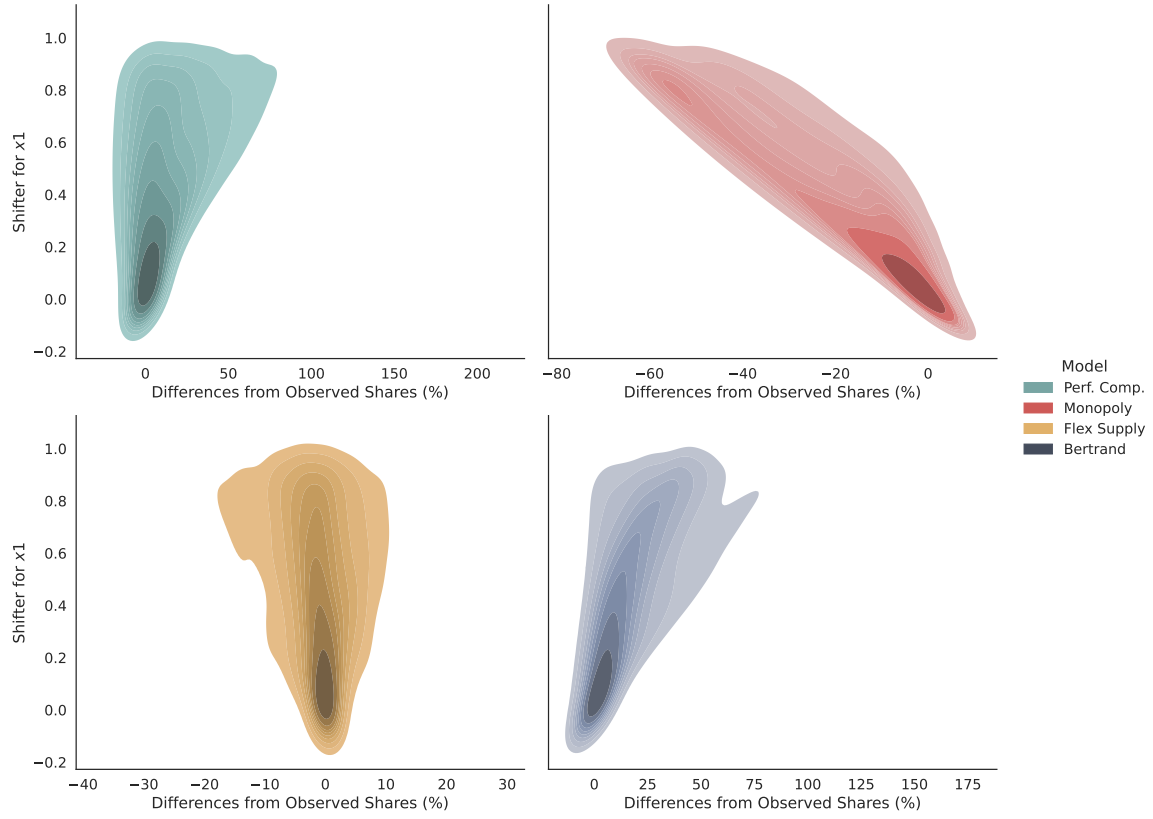
Notes: Counterfactual price predictions as x_1 increases by varying factors, Bertrand DGP. Flexible model: $|h| = 3$, $T = 10,000$.

FIGURE S.3.8: Cost Shifter Regulation, Bertrand DGP



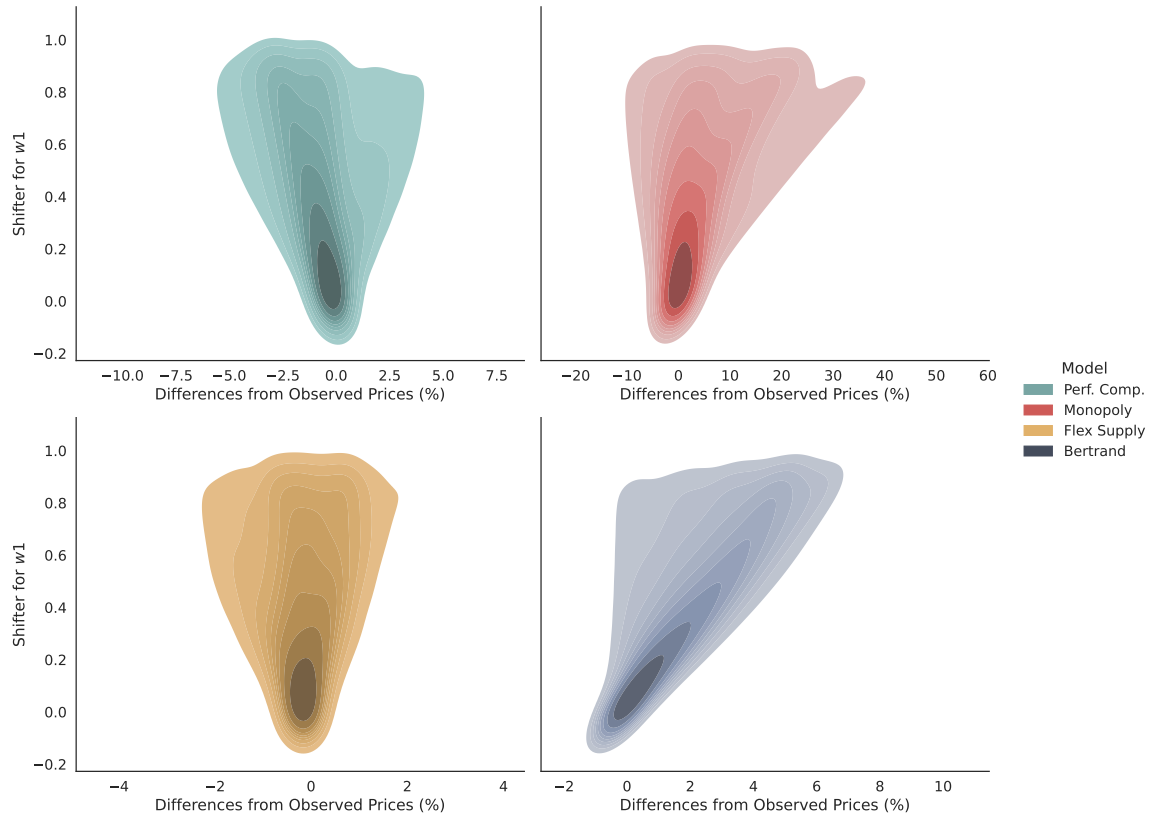
Notes: Counterfactual price predictions as w_1 increases by varying factors, Bertrand DGP. Flexible model: $|h| = 3$, $T = 10,000$.

FIGURE S.3.9: Product Characteristics Regulation, Profit-Weight DGP



Notes: Counterfactual price predictions as x_1 increases by varying factors, profit-weight DGP ($\kappa = 0.5$). Flexible model: $|h| = 3$, $T = 10,000$.

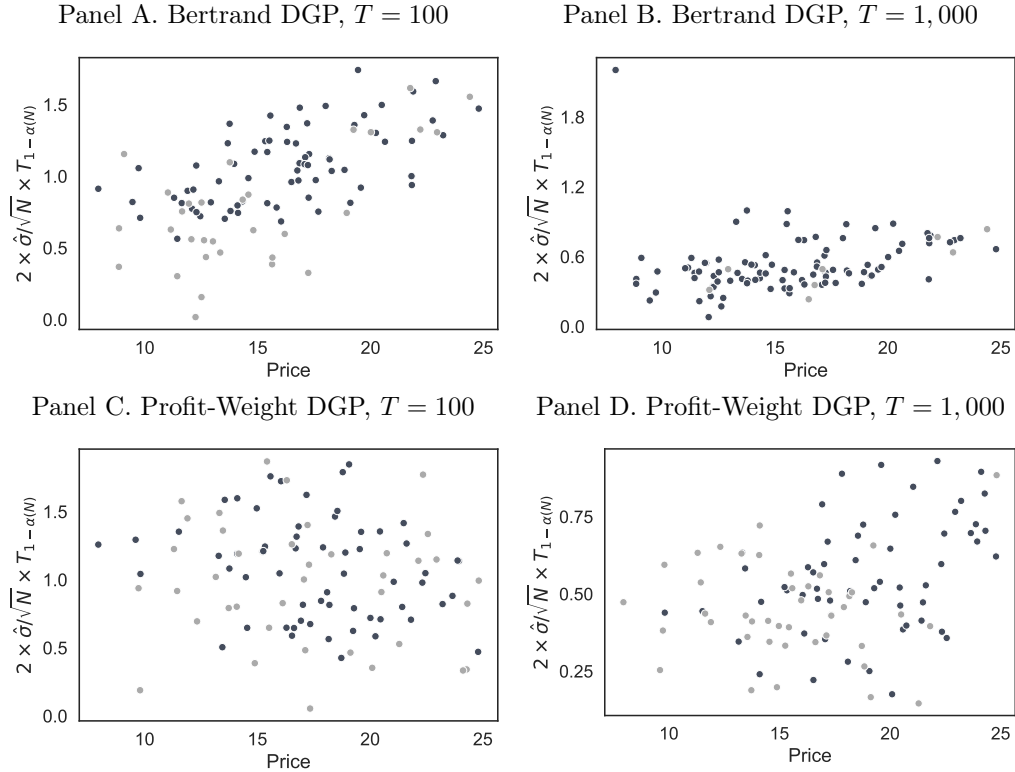
FIGURE S.3.10: Cost Shifter Regulation, Profit-Weight DGP



Notes: Counterfactual price predictions as cost shifters increase by varying factors, profit-weight DGP ($\kappa = 0.5$). Flexible model: $|h| = 3$, $T = 10,000$.

S.3.3 Additional Inference Results

FIGURE S.3.11: Inference on Counterfactual Merger Simulation Prices



Panel E. Inference Coverage

	$T = 100$		$T = 1,000$	
	A. Bertrand	B. Profit-Weight	C. Bertrand	D. Profit-Weight
Within	69%	61%	92%	58%
Outside	31%	39%	8%	42%

Notes: Bonferroni-corrected confidence interval widths for counterfactual prices following a product exit. Flexible model: $|h| = 20$, D_t excluded.

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